

# Cognitive Ability, Attentional Regulation, and Academic Achievement

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## Abstract

Cognitive and noncognitive skills both predict student achievement, but how they combine at the margin remains unsettled. Using the Growing Up in Ireland longitudinal study, I examine how attentional regulation and cognitive ability - both measured at age 13 - jointly predict Junior Certificate achievement in Maths and English, conditioning on age-9 baselines. The cognition  $\times$  attentional regulation interaction is negative and significant in both subjects: the marginal return to cognitive ability is larger for students with weaker behavioural regulation than for those with stronger regulation. The result survives multiple-testing correction, inverse-probability reweighting for sample selection, omitted-variable-bias bounds, and design-effect-adjusted inference, and it reappears when attentional regulation is rated by the child's class teacher four years earlier - the strongest available check against parent-specific reporting style. A translog specification points in the same direction but is too sensitive to evaluation point to anchor a nonlinear claim. Re-estimating the model with a broader, trait-level personality proxy reported by the same informant at the same age yields a weak and imprecise interaction. The conclusion about how cognitive and noncognitive skills combine therefore depends on the behavioural content of the noncognitive measure, not only on whether a noncognitive input is included. Cognitive ability and attentional regulation appear to function as compensating inputs in predicting later academic achievement.

Keywords: cognitive skills, noncognitive skills, educational production, academic performance, gender differences, human capital

JEL Codes: I21, I24, J24, C31

# 1 Introduction

Cognitive and noncognitive skills both predict student achievement, but how they combine at the margin remains unsettled (Almlund et al., 2011; Humphries & Kosse, 2017). Three possibilities have different implications for how achievement is produced: the skills may reinforce one another (so balanced development matters most), compensate for one another (so strength on one margin can offset weakness on the other), or contribute additively (so each can be targeted in isolation). Existing work establishes that both types of skill matter, but it has not asked whether the estimated interaction between them is sensitive to how noncognitive skill is measured - a gap this chapter fills by showing that the answer changes substantially across commonly used constructs.

I examine how attentional regulation and cognitive ability jointly predict achievement on Ireland’s lower-secondary school-leaving examination, the Junior Certificate, using the Growing Up in Ireland (GUI) longitudinal study. The main specification is a linear interaction model for Maths and English, with cognitive ability and a measure of attentional regulation both observed at age 13, baseline controls from age 9, and robust standard errors. To test whether the substantive answer depends on how noncognitive skill is measured, I re-estimate the models replacing attentional regulation with a broader, trait-level measure of conscientiousness, reported by the same informant at the same age. As a further check against functional-form dependence, I estimate translog specifications that allow the interaction to take a nonlinear shape.

The main result is that cognitive ability and attentional regulation interact negatively in both subjects: the marginal return to cognitive ability is larger for students weaker on attentional regulation than for those already strong on it. In the preferred specification, the interaction coefficient is about 0.05 in Maths and 0.06 in English. Moving from one standard deviation below the mean in attentional regulation to one above it reduces the estimated return to cognitive ability by roughly 0.10 log points in Maths and 0.12 log points in English. The two inputs therefore compensate at the margin rather than reinforce one another. This is a different question from the dynamic complementarities emphasised in the skill-formation literature, which asks how earlier skills raise the return to later investment; here the question is how two contemporaneously observed skills combine in predicting a later outcome. Under the broader conscientiousness measure, the corresponding interaction is close to zero and statistically imprecise. The contrast is necessary: the estimated interaction depends not just on whether a noncognitive skill is included, but on which construct is measured.

I place most weight on the linear interaction evidence. The negative interaction reappears when attentional regulation is measured by a teacher rather than a parent, using a cross-informant version of the same construct collected four years earlier - the

strongest available check against the concern that the headline result reflects parent-specific reporting style. The interaction also survives multiple-testing correction, reweighting for sample selection, omitted-variable-bias bounds, and inflated standard errors under plausible school-level clustering. The translog exercise serves as a secondary check: it points in the same direction, but its nonlinear summaries are too specification-sensitive to anchor the main claim.

Ireland is useful for this analysis mainly because the data allow the question to be asked cleanly. The Growing Up in Ireland study follows the same cohort from middle childhood into adolescence and links a rich cognitive battery, distinct noncognitive measures, and later high-stakes examination outcomes. Within a single cohort and institutional setting, this makes it possible to ask both how cognitive and noncognitive skills combine and whether the estimated combination changes when the noncognitive construct changes.

Throughout, I interpret the estimates as conditional associations in the analysis sample. The design relies on temporal ordering, cross-student variation in observed inputs, and a rich set of controls, but not on exogenous treatment assignment. The contribution is therefore descriptive rather than causal: evidence on how cognitive ability and attentional regulation combine in predicting later achievement, and on how far that conclusion depends on whether the noncognitive measure captures classroom behaviour or broader personality - a distinction that has bearing on how interaction estimates in the skill-formation literature should be read.

The next subsection reviews the related literature. Section 2 describes the data and measurement instruments. Section 3 presents the linear interaction models and the main results. Section 4 reports the translog sensitivity exercise. Section 5 presents the inference and robustness battery, including the cross-informant check. Section 6 concludes. Detailed derivations and additional diagnostics appear in the appendices.

## 1.1 Related literature

Three strands of literature motivate this chapter. The first concerns the specification of educational production functions. Todd and Wolpin (2003) show that estimated skill effects depend on what is treated as contemporaneous input, what is absorbed into lagged stocks, and how prior achievement is handled. That framework bears directly on the design here: the chapter studies a contemporaneous interaction between cognitive ability and attentional regulation measured at age 13, with age 9 measures as baseline controls. The focus is on how two observed skills combine in predicting a later educational outcome under a transparent conditional-association design.

The second strand concerns the role of noncognitive skills in achievement. A large literature shows that traits such as self-regulation, perseverance, and conscientiousness predict educational attainment and broader life outcomes alongside measured cognitive

ability (Almlund et al., 2011; Heckman et al., 2006; Kautz et al., 2014; Poropat, 2009). Related work shows that noncognitive skills are consequential within school settings and in long-run labour-market outcomes: teacher effects on noncognitive outcomes predict later success (Jackson, 2018), and social skills have become increasingly rewarded in the labour market (Deming, 2017). Much of this literature, however, treats noncognitive skills as additive predictors and says less about how they operate as interacting inputs within the same educational production process. How cognitive and noncognitive skills combine within that process is the question this chapter takes up.

A related question arises from the skill-formation literature. In dynamic models, early skills can raise the return to later investment in other skills (Borghans et al., 2008; Cunha & Heckman, 2008) - a claim about cross-productivity over time. It does not pin down the sign of the contemporaneous interaction between two observed skills measured at a single developmental stage. A negative interaction at age 13 is therefore not in contradiction with positive dynamic complementarities earlier in childhood; the two objects are different.

Gil-Hernández (2021) is the closest paper to the present chapter. Using German panel data on the transition into secondary schooling, he studies whether the educational return to conscientiousness is larger for cognitively weaker students, especially among those from high-SES families, and finds evidence consistent with a compensatory pattern: cognitively weaker high-SES students receive higher returns to conscientiousness. That finding is directionally consistent with the negative interaction estimated here, though on a different outcome, sample, and noncognitive construct. What separates the two papers is the question being asked.

Gil-Hernández (2021) studies whether cognitive and noncognitive skills interact rather than treating them as separate predictors; this chapter asks whether the estimated interaction itself shifts when the noncognitive construct varies - holding outcome, cohort, and modelling framework fixed while varying the behavioural content of the noncognitive measure.

The third strand concerns measurement. Noncognitive constructs span broad personality traits, behavioural indicators, and economic preferences, with no single consensus instrument (Borghans et al., 2008; Humphries & Kosse, 2017). This is consequential here because the chapter contrasts two measures that differ not only in content but also in psychometric design and intended use. The Strengths and Difficulties Questionnaire (SDQ) is a multi-informant behavioural screening instrument designed for children and adolescents; Goodman (2001) reports satisfactory psychometric performance overall and good discrimination for the hyperactivity-inattention subscale, which is closest to the attentional-regulation margin studied here. The SDQ measure used in the chapter - termed Focused Behaviour - is the reverse-coded hyperactivity-inattention subscale, so this psychometric evidence applies directly. Reviews of the parent- and teacher-report

SDQ find that the hyperactivity scale outperforms several other subscales on reliability and validity, especially the peer problems and conduct problems scales (Stone et al., 2010). The Ten-Item Personality Inventory (TIPI), by contrast, was designed as an ultra-brief Big Five measure for settings where personality is not the primary object of measurement. Gosling et al. (2003) show that it has adequate convergent validity and test-retest reliability for a ten-item instrument, but state clearly that it is psychometrically inferior to longer Big Five inventories and is best used when brevity is a high priority.

Beyond the three strands above, a separate literature motivates the chapter's secondary gender analysis. Girls tend to display stronger self-discipline and classroom behaviour than boys at similar levels of measured cognitive skill, and those differences may shape how cognitive and noncognitive skills combine in achievement (Bertrand & Pan, 2013; Duckworth & Seligman, 2005). Gender heterogeneity is therefore treated as a secondary dimension of the analysis rather than the chapter's main contribution. Taken together, the literature has established that cognitive and noncognitive skills both contribute to educational success, suggested that their interaction may be compensatory or reinforcing, and documented that noncognitive measures differ in both behavioural content and psychometric properties. What remains less clear is whether the estimated interaction between cognitive ability and noncognitive skill is stable across plausible measures of the noncognitive margin. That is the gap this chapter addresses, with implications for how interaction estimates in the skill-formation literature should be interpreted. The SDQ and TIPI questionnaires are reproduced in Appendices 6.6 and 6.5.

## 2 Data and measurement

This section describes the data and the construction of the analysis variables. Subsection 2.1 introduces the GUI study, the analysis sample, and the descriptive statistics. Subsection 2.2 describes the outcome, the cognitive and noncognitive inputs, and the controls.

### 2.1 Data

**The Growing Up in Ireland study.** The data come from Growing Up in Ireland (GUI), the first nationally representative longitudinal survey of children in Ireland. The Cohort '98 sample comprises 8,568 children born between November 1997 and October 1998, interviewed at ages 9 (Wave 1, 2007–08), 13 (Wave 2, 2011–12), and 17/18 (Wave 3, 2015–16). Retention was 87.8% at Wave 2 and 72.6% at Wave 3. Table 1 summarises the timing structure.

| Event                            | Date            | Age (in years) | Variables of interest                                                              |
|----------------------------------|-----------------|----------------|------------------------------------------------------------------------------------|
| Study-child is born              | Nov/97 – Oct/98 | 0              |                                                                                    |
| Wave 2 data collection           | Aug/11 – Mar/12 | 13             | Independent variables:<br>Cognitive variables,<br>SDQ and TIPI scales,<br>controls |
| Study-child sits the Junior Cert | Jun/13 – Jun/14 | 15–16          |                                                                                    |
| Wave 3 data collection           | Apr/15 – Aug/16 | 17–18          | Dependent variables:<br>Junior Cert scores in<br>Maths and English                 |

Table 1: Timing structure of the analysis in GUI Cohort '98. Predictors are measured at Wave 2 (age 13), while Junior Certificate outcomes (sat at ages 15–16) are reported at Wave 3 (ages 17–18), so that all predictors precede the outcome by two to three years.

**Sample construction.** For estimation I use a complete-case longitudinal subsample in which individuals have non-missing values for the variables required by each model block. This keeps sample composition fixed within each comparison, so model differences cannot be confused with sample differences. The trade-off is external validity: the estimand is an association in this analytically consistent subsample, narrower in scope than a population average over the full GUI cohort.

Five distinct sample sizes appear in the analysis. Table 2 summarises the flow.

**Selection and attrition.** The retained analytical sample is positively selected on prior achievement and family resources: retained children scored higher on Wave 1 reading ( $d = +0.35$ ) and maths ( $d = +0.34$ ) and came from higher-income families ( $d = +0.23$ ). Within the analytical sample, the further restriction to complete controls ( $N \approx 3,786$ ) is

Table 2: Sample flow from the original GUI cohort to model-specific estimation samples. Survey attrition drives the first-stage reduction; later reductions are analyst-imposed complete-case requirements. The OLS TIPI panel is slightly smaller than the OLS SDQ panel because the TIPI panel requires non-missing values for all five traits and three observations have at least one missing item at the Model 4 estimation stage. The translog pipeline (Section 4.1) uses only the focal noncognitive variable per instrument and is harmonised across instruments to a single common sample of  $N = 5,631$ . The 163 observations with missing gender are retained in the pooled main specifications but dropped from gender-subgroup analyses.

| Stage                                                                            | $N$   | % of Wave 1 |
|----------------------------------------------------------------------------------|-------|-------------|
| GUI '98 Cohort, Wave 1 (age 9)                                                   | 8,568 | 100.0       |
| Retained at Wave 2 (age 13)                                                      | 7,525 | 87.8        |
| Retained at Wave 3 (age 17–18)                                                   | 6,216 | 72.6        |
| <i>Linear-model (OLS) SDQ panel</i>                                              |       |             |
| Complete case: cognition + 4 SDQ subscales + JC outcomes                         | 5,631 | 65.7        |
| + Complete SES and school controls (Model 3)                                     | 3,786 | 44.2        |
| + Wave 1 baselines (Model 4, preferred)                                          | 3,711 | 43.3        |
| <i>Linear-model (OLS) TIPI panel</i>                                             |       |             |
| Complete case: cognition + 5 TIPI traits + JC outcomes                           | 5,625 | 65.7        |
| + Complete SES and school controls (Model 3)                                     | 3,783 | 44.1        |
| + Wave 1 baselines (Model 4, preferred)                                          | 3,708 | 43.3        |
| <i>Translog common sample (SDQ Focused Behaviour and TIPI Conscientiousness)</i> |       |             |
| Complete case: cognition + focal noncognitive + JC outcomes                      | 5,631 | 65.7        |
| <i>Gender subgroups (translog common sample)</i>                                 |       |             |
| Boys                                                                             | 2,667 | —           |
| Girls                                                                            | 2,801 | —           |
| Gender missing                                                                   | 163   | —           |

only weakly patterned on observables (logit pseudo- $R^2 = 0.014$ ). Full baseline-balance diagnostics, IPW sensitivity, and sample-flow details are reported in Appendix 6.8. The direction of this selection - compression of the lower tail of the skill distribution - would be expected to attenuate any interaction heterogeneity at lower skill levels rather than inflate it, an interpretation point I return to in Section 5.

**Descriptive statistics.** Table 3 reports descriptive statistics for outcomes, skill inputs, and controls in the translog common sample of  $N = 5,631$ .

Table 3: Descriptive statistics for outcomes, skill inputs, and controls in the translog common sample ( $N = 5,631$ ). Outcomes are Junior Certificate OPS points, cognition is based on Wave 2 test components, and SDQ variables are recoded so higher values indicate stronger positive traits.

| Variable                                          | Mean   | Std. Dev. | Min   | Max    | N    |
|---------------------------------------------------|--------|-----------|-------|--------|------|
| <b>Dependent variables</b>                        |        |           |       |        |      |
| Maths points (Junior Cert)                        | 9.60   | 1.74      | 2.00  | 12.00  | 5631 |
| English points (Junior Cert)                      | 10.15  | 1.34      | 5.00  | 12.00  | 5631 |
| <b>Independent variables: Cognition</b>           |        |           |       |        |      |
| Drumcondra Verbal Reasoning (% correct answers)   | 64.89  | 21.92     | 0.00  | 100.00 | 5631 |
| Drumcondra Numerical Ability (% correct answers)  | 55.05  | 22.53     | 0.00  | 100.00 | 5631 |
| Matrices (BAS)                                    | 116.68 | 18.03     | 10.00 | 161.00 | 5631 |
| Cognition PC1 (raw first component)               | 0.00   | 1.37      | -5.84 | 3.33   | 5631 |
| Cognition (shifted, mean = 100)                   | 100.00 | 15.00     | 36.25 | 136.40 | 5631 |
| <b>Independent variables: Noncognition (SDQ)</b>  |        |           |       |        |      |
| Emotional resilience                              | 8.29   | 1.87      | 0.00  | 10.00  | 5631 |
| Good conduct                                      | 8.97   | 1.31      | 0.00  | 10.00  | 5631 |
| Focused behaviour                                 | 7.56   | 2.26      | 0.00  | 10.00  | 5631 |
| Positive peer relationships                       | 8.96   | 1.41      | 0.00  | 10.00  | 5631 |
| <b>Independent variables: Noncognition (TIPI)</b> |        |           |       |        |      |
| Agreeable                                         | 5.01   | 1.95      | 0.50  | 7.00   | 5631 |
| Conscientious                                     | 4.33   | 2.07      | 0.50  | 7.00   | 5631 |
| Emotional stability                               | 4.40   | 1.99      | 0.50  | 7.00   | 5631 |
| Extravert                                         | 3.98   | 1.98      | 0.50  | 7.00   | 5629 |
| Openness                                          | 4.73   | 1.83      | 0.50  | 7.00   | 5627 |
| <b>Controls (SES characteristics)</b>             |        |           |       |        |      |
| Gender (Male = 1)                                 | 0.49   | 0.50      | 0.00  | 1.00   | 5468 |
| Primary caregiver education level                 | 3.97   | 1.24      | 1.00  | 6.00   | 5631 |
| Secondary caregiver education level               | 3.86   | 1.36      | 1.00  | 6.00   | 4440 |
| Income quintile (equivalised)                     | 3.33   | 1.39      | 1.00  | 5.00   | 5241 |
| <b>Controls (School characteristics, binary)</b>  |        |           |       |        |      |
| DEIS                                              | 0.12   | 0.33      | 0.00  | 1.00   | 5452 |
| Fee-paying                                        | 0.10   | 0.30      | 0.00  | 1.00   | 5452 |
| Mixed-school                                      | 0.54   | 0.50      | 0.00  | 1.00   | 5317 |

Note: The original SDQ scales (0–10) have been inverted so that higher scores indicate stronger positive traits. “Cognition PC1 (raw first component)” is the first principal component of the three standardised cognitive subtests; the linear OLS models use a unit-standard-deviation rescaling of this component. “Cognition (shifted, mean = 100)” is a separate positive rescaling of the same component, used in the translog specifications. The TIPI sample minima of 0.50 reflect the W2 derived-variable construction issue documented in the noncognitive measurement paragraph; see Appendix 6.9, Table 20, for a sensitivity check.



## 2.2 Measurement

This subsection describes how each variable is constructed from the GUI data: the Junior Certificate outcome, the cognitive composite, the two noncognitive instruments, and the controls. The estimation rescaling conventions are noted at the end.

**Outcome.** The outcome is performance on the Junior Certificate (JC), a national examination taken by most Irish students at ages 15–16, covering mandatory and elective subjects across levels. I map reported grades to the standardised 12-point Overall Performance Scale (OPS), following Sofroniou et al. (2000), so achievement is comparable across subjects and levels. Descriptive statistics are reported in raw OPS points; the main linear specifications use  $\log(\text{OPS})$  as the dependent variable, and the z-score appendix check verifies that the interaction sign does not depend on this transformation.

**Cognitive input.** Cognitive ability is measured by three Wave 2 tests: the Drumcondra Verbal Reasoning and Numerical Ability assessments, and the British Ability Scales (BAS) Matrices test of non-verbal reasoning. I combine these into a single principal component and use it throughout as the cognitive input for both Maths and English outcomes. Using a single cognitive composite follows standard practice in the education production-function literature, where multiple test scores serve as indicators of one latent cognitive factor (Attanasio et al., 2020; Cunha et al., 2010).<sup>1</sup> The first principal component has mean zero by construction and sample standard deviation approximately 1.37 (Table 3).

**Noncognitive inputs.** Noncognitive skills are measured at Wave 2 using two alternative instruments, both reported by the primary caregiver.<sup>2</sup> Estimating separate specifications using each instrument allows direct comparison of how the choice of noncognitive measure shapes the recovered interaction.

The Strengths and Difficulties Questionnaire (SDQ; Goodman, 1997) is a 20-item instrument measuring behavioural and emotional characteristics through four difficulty subscales (each scored 0–10): Emotional Symptoms, Conduct Problems, Hyperactivity/Inattention, and Peer-relationship Problems. I invert the scales so that higher values indicate more positive traits, renaming them Emotional Resilience, Good Conduct, Focused Behaviour, and Positive Peer Relationships. The focal SDQ measure is Focused Behaviour, the reverse-coded Hyperactivity/Inattention subscale. The SDQ captures

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<sup>1</sup>Agostinelli and Wiswall (2025) instead estimate separate production functions for maths and reading latent skills, but the changing test batteries across GUI waves do not straightforwardly support that design here.

<sup>2</sup>In the GUI '98 Cohort, the primary caregiver is self-identified as the person who provides most care and knows most about the child. In the analytical sample, 97.7% of primary caregivers are the biological mother. “Primary caregiver” reports here should therefore be understood as maternal assessments.

school-proximal behavioural regulation: behaviours directly observable in classroom and study settings and therefore tightly linked to school performance.

The Ten Item Personality Inventory (TIPI; Gosling et al., 2003) is a brief instrument assessing the Big Five personality dimensions (Openness, Agreeableness, Conscientiousness, Extraversion, and Emotional Stability) on a seven-point scale. I use Wave 2 primary-caregiver reports, collected at the same wave and from the same informant as the SDQ and cognitive measures. The TIPI captures broader personality traits less directly tied to classroom behaviour than the SDQ. A known data-construction issue affects a small number of Wave 2 TIPI cases: the GUI-released W2 derived aggregates exhibit a sample minimum of 0.5 inconsistent with documentation, affecting 27–143 cases per trait. Item-level W2 TIPI variables are not in the analytical extract and the affected cases cannot be reconstructed. A direct sensitivity check trimming the visibly contaminated cases is reported in Appendix 6.9, Table 20; the focal interaction remains weak and imprecise after trimming.

Full questionnaires and scoring are reproduced in Appendices 6.5 and 6.6.

**Controls.** Control variables fall into two blocks. The socioeconomic block includes gender, parental education (primary and secondary caregiver), and equivalised income quintiles (Davis-Kean, 2005; Sirin, 2005). The school block includes mixed-gender status, Delivering Equality of Opportunity in Schools (DEIS) designation, and fee-paying status (OECD, 2012; Pahlke et al., 2014; Smyth et al., 2015).

**Estimation conventions.** For the linear regressions, all skill inputs are rescaled to mean zero and unit standard deviation, so coefficients represent the change in log achievement associated with a one-standard-deviation increase in the input. For the translog specifications, the cognitive composite is mapped through a separate strictly positive rescaling - to mean 100 and standard deviation 15, following the psychometric convention - because translog estimation requires strictly positive inputs.

### 3 Linear production function

#### 3.1 Specification

I begin with a linear production function that relates logged Junior Certificate achievement to cognitive ability, noncognitive measures, and their interactions. The interaction terms are the central objects of interest. They capture whether the marginal association of one input with achievement varies with the level of the other: a negative interaction coefficient implies a negative cross-partial in the estimated conditional mean, so the marginal association of cognition with achievement declines as noncognitive skill increases, and vice versa.

The specification is:

$$\begin{aligned} \log(\text{PointsJC}_{i,l}) = & \beta_{0l} + \beta_{Cl} \cdot \text{Cognition}_i + \sum_{j=1}^J \beta_{Njl} \cdot \text{NonCog}_{i,j} \\ & + \sum_{j=1}^J \gamma_{jl} \cdot (\text{Cognition}_i \cdot \text{NonCog}_{i,j}) + \boldsymbol{\delta}'_l \cdot \mathbf{X}_i + \varepsilon_{i,l} \end{aligned} \quad (1)$$

where  $i$  indexes individuals,  $l$  subjects (Maths, English) - estimated separately - and  $j$  noncognitive dimensions. The vector  $\mathbf{X}_i$  contains the socioeconomic and school controls described in Section 2.

I estimate four nested models per subject: Model 1 includes only cognitive and noncognitive inputs; Model 2 adds socioeconomic and school controls; Model 3 adds cognition  $\times$  noncognition interactions; Model 4 - the preferred specification - adds Wave 1 reading, maths, and SDQ baseline measures. Models 1–3 are reported in Tables 4 and 5; the preferred Model 4 coefficients appear in Table 6, with Figures 1 and 2 as visual companions. The with-versus-without-Wave 1 comparison that anchors the robustness claim is reported in Table 11 in Section 5. The same specification is estimated for both the SDQ and TIPI noncognitive sets, allowing direct comparison.

#### 3.2 Results

Table 4 reports the primary SDQ results and Table 5 reports the TIPI contrast. I discuss the key patterns below, placing most weight on the preferred Model 4 coefficients in Table 6.

Three findings organise the discussion: cognition is the dominant first-order predictor; among the SDQ subscales, attentional regulation is the strongest noncognitive predictor; and the focal cognition  $\times$  noncognition interaction is negative under the SDQ but small and imprecise under the TIPI.

Table 4: OLS associations between cognition, SDQ behavioural characteristics, and Junior Certificate achievement in Maths and English. Dependent variables are log OPS points (Maths 2–12; English 5–12); predictors are standardised; Model (1) includes cognition and SDQ only, Model (2) adds socioeconomic and school controls, and Model (3) adds cognition-trait interactions. Standard errors are reported in parentheses. The preferred baseline-control Model 4 coefficients are reported in Table 6.

| Model:                      | Maths                 |                       |                        | English               |                        |                        |
|-----------------------------|-----------------------|-----------------------|------------------------|-----------------------|------------------------|------------------------|
|                             | (1)                   | (2)                   | (3)                    | (1)                   | (2)                    | (3)                    |
| <i>Variables</i>            |                       |                       |                        |                       |                        |                        |
| Constant                    | 8.5629***<br>(0.1085) | 8.7291***<br>(0.1314) | 8.8320***<br>(0.1294)  | 9.4481***<br>(0.0992) | 9.7286***<br>(0.1150)  | 9.8366***<br>(0.1123)  |
| Cognition                   | 0.7763***<br>(0.0135) | 0.6675***<br>(0.0172) | 0.9811***<br>(0.0846)  | 0.4595***<br>(0.0124) | 0.4135***<br>(0.0149)  | 0.7233***<br>(0.0807)  |
| Emotional Resilience        | 0.0507**<br>(0.0198)  | 0.0544**<br>(0.0233)  | 0.0444*<br>(0.0228)    | −0.0258<br>(0.0179)   | 0.0092<br>(0.0206)     | −0.0007<br>(0.0203)    |
| Good Conduct                | 0.0832***<br>(0.0207) | 0.0693***<br>(0.0250) | 0.0714***<br>(0.0246)  | 0.0060<br>(0.0191)    | −0.0201<br>(0.0221)    | −0.0161<br>(0.0215)    |
| Focused Behaviour           | 0.2447***<br>(0.0210) | 0.2283***<br>(0.0249) | 0.2175***<br>(0.0247)  | 0.2440***<br>(0.0187) | 0.1906***<br>(0.0218)  | 0.1782***<br>(0.0213)  |
| Positive Peer Relationships | 0.0197<br>(0.0181)    | 0.0002<br>(0.0211)    | 0.0046<br>(0.0224)     | 0.0704***<br>(0.0170) | 0.0502**<br>(0.0199)   | 0.0565***<br>(0.0208)  |
| Male                        |                       | −0.0707*<br>(0.0398)  | −0.0702*<br>(0.0397)   |                       | −0.4019***<br>(0.0342) | −0.3996***<br>(0.0339) |
| Cognition × E.R.            |                       |                       | −0.0283*<br>(0.0163)   |                       |                        | −0.0148<br>(0.0154)    |
| Cognition × G.C.            |                       |                       | 0.0122<br>(0.0181)     |                       |                        | 0.0247<br>(0.0180)     |
| Cognition × F.B.            |                       |                       | −0.0567***<br>(0.0174) |                       |                        | −0.0737***<br>(0.0171) |
| Cognition × P.P.R.          |                       |                       | −0.0158<br>(0.0144)    |                       |                        | −0.0261*<br>(0.0147)   |
| <i>Controls</i>             |                       |                       |                        |                       |                        |                        |
| SES                         | No                    | Yes                   | Yes                    | No                    | Yes                    | Yes                    |
| School                      | No                    | Yes                   | Yes                    | No                    | Yes                    | Yes                    |
| <i>Fit statistics</i>       |                       |                       |                        |                       |                        |                        |
| Observations                | 5,631                 | 3,786                 | 3,786                  | 5,631                 | 3,786                  | 3,786                  |
| R <sup>2</sup>              | 0.47270               | 0.49904               | 0.50214                | 0.30700               | 0.34584                | 0.35277                |
| Adjusted R <sup>2</sup>     | 0.47223               | 0.49744               | 0.50002                | 0.30639               | 0.34376                | 0.35003                |

*Heteroskedasticity-robust (HC1) standard errors in parentheses.* \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

*E.R.: Emotional Resilience, G.C.: Good Conduct, F.B.: Focused Behaviour,*

*P.P.R.: Positive Peer Relationships*

Note: Regressors are standardised (mean 0, SD 1). Coefficients are reported in log-point units per 1 SD increase in the predictor.

**First-order patterns.** Cognitive ability carries the dominant first-order association with achievement in every specification: across both tables, its coefficient is larger and more precisely estimated than any noncognitive main effect. Among the SDQ subscales, Focused Behaviour is the strongest noncognitive predictor. Its Model 1 Maths coefficient

Table 5: OLS associations between cognition, TIPI personality traits, and Junior Certificate achievement in Maths and English. Dependent variables are log OPS points (Maths 2–12; English 5–12); predictors are standardised; Model (1) includes cognition and TIPI only, Model (2) adds socioeconomic and school controls, and Model (3) adds cognition-trait interactions. Standard errors are reported in parentheses. The preferred baseline-control Model 4 coefficients are reported in Table 6.

| Model:                  | Maths                  |                        |                        | English               |                        |                        |
|-------------------------|------------------------|------------------------|------------------------|-----------------------|------------------------|------------------------|
|                         | (1)                    | (2)                    | (3)                    | (1)                   | (2)                    | (3)                    |
| <i>Variables</i>        |                        |                        |                        |                       |                        |                        |
| Constant                | 9.1703***<br>(0.0480)  | 9.4124***<br>(0.0633)  | 9.3992***<br>(0.0638)  | 9.9510***<br>(0.0423) | 10.2783***<br>(0.0533) | 10.2705***<br>(0.0538) |
| Cognition               | 0.8280***<br>(0.0129)  | 0.7130***<br>(0.0167)  | 0.8777***<br>(0.0428)  | 0.4993***<br>(0.0122) | 0.4437***<br>(0.0148)  | 0.5300***<br>(0.0399)  |
| Agreeableness           | 0.0090<br>(0.0182)     | 0.0286<br>(0.0216)     | 0.0306<br>(0.0220)     | 0.0272*<br>(0.0160)   | 0.0348*<br>(0.0182)    | 0.0365**<br>(0.0186)   |
| Conscientiousness       | 0.1609***<br>(0.0181)  | 0.1349***<br>(0.0208)  | 0.1397***<br>(0.0215)  | 0.0900***<br>(0.0155) | 0.0751***<br>(0.0176)  | 0.0791***<br>(0.0182)  |
| Emotional stability     | 0.0661***<br>(0.0182)  | 0.0474**<br>(0.0211)   | 0.0567***<br>(0.0219)  | 0.0135<br>(0.0160)    | 0.0046<br>(0.0178)     | 0.0087<br>(0.0185)     |
| Extraversion            | -0.0177<br>(0.0175)    | -0.0206<br>(0.0201)    | -0.0185<br>(0.0208)    | 0.0164<br>(0.0151)    | 0.0114<br>(0.0168)     | 0.0135<br>(0.0174)     |
| Openness                | -0.0752***<br>(0.0175) | -0.0225<br>(0.0203)    | -0.0238<br>(0.0209)    | 0.0063<br>(0.0158)    | 0.0113<br>(0.0176)     | 0.0141<br>(0.0182)     |
| Male                    |                        | -0.1142***<br>(0.0397) | -0.1160***<br>(0.0397) |                       | -0.4452***<br>(0.0344) | -0.4440***<br>(0.0344) |
| Cognition × Agree.      |                        |                        | -0.0168<br>(0.0166)    |                       |                        | -0.0118<br>(0.0151)    |
| Cognition × Consc.      |                        |                        | -0.0168<br>(0.0159)    |                       |                        | -0.0176<br>(0.0146)    |
| Cognition × Emot.       |                        |                        | -0.0669***<br>(0.0159) |                       |                        | -0.0264*<br>(0.0148)   |
| Cognition × Extra.      |                        |                        | 0.0011<br>(0.0155)     |                       |                        | -0.0063<br>(0.0139)    |
| Cognition × Open.       |                        |                        | 0.0074<br>(0.0162)     |                       |                        | -0.0240*<br>(0.0142)   |
| <i>Controls</i>         |                        |                        |                        |                       |                        |                        |
| SES                     | No                     | Yes                    | Yes                    | No                    | Yes                    | Yes                    |
| School                  | No                     | Yes                    | Yes                    | No                    | Yes                    | Yes                    |
| <i>Fit statistics</i>   |                        |                        |                        |                       |                        |                        |
| Observations            | 5,625                  | 3,783                  | 3,783                  | 5,625                 | 3,783                  | 3,783                  |
| R <sup>2</sup>          | 0.45483                | 0.48360                | 0.48732                | 0.27840               | 0.33049                | 0.33361                |
| Adjusted R <sup>2</sup> | 0.45425                | 0.48182                | 0.48487                | 0.27763               | 0.32818                | 0.33042                |

*Heteroskedasticity-robust (HC1) standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .*

*Agree.: Agreeableness, Consc.: Conscientiousness, Emot.: Emotional stability,*

*Extra.: Extraversion, Open.: Openness*

Note: Regressors are standardised (mean 0, SD 1). Coefficients are reported in log-point units per 1 SD increase in the predictor.

(0.245) is roughly three to four times the size of the next strongest behavioural margin (Good Conduct at 0.083, Emotional Resilience at 0.051), and it is also the most stable across Models 1–3. This first-order gap helps justify the focus on attentional regulation when turning to interactions, ahead of the other behavioural margins.

**The focal interaction.** Under the preferred baseline-control specification, the Cognition  $\times$  Focused Behaviour interaction is  $-0.052$  in Maths and  $-0.062$  in English (Table 6). Both estimates are statistically significant at the 1% level and confirm the predominantly negative interaction pattern visible in the stepped Models 1–3. The marginal return to one input is therefore smaller when the other is already high.<sup>3</sup> The sign runs against the positive dynamic cross-productivity often emphasised in early-childhood skill-formation models, but the objects differ: this is a contemporaneous interaction at age 13, not a claim about how earlier investments raise later returns.

**Construct sensitivity.** Under the broader personality proxy, the corresponding Cognition  $\times$  Conscientiousness interactions are much smaller ( $-0.014$  in Maths and  $-0.015$  in English) and statistically imprecise. This is the chapter’s central methodological finding: the recovered joint-skill relationship depends on which noncognitive construct is used.

The TIPI grid is not uniformly null, however. The Cognition  $\times$  Emotional Stability interaction in Maths is  $-0.067$  and significant at the 1% level - larger in absolute size than the focal SDQ result. I do not treat this as the focal interaction for two reasons. First, the SDQ result is pre-specified by the chapter’s construct logic, which targets school-proximal behavioural regulation rather than personality stability. Second, the TIPI Emotional Stability interaction does not replicate cleanly in English ( $-0.026$ , marginal at the 10% level), whereas the SDQ Focused Behaviour result is stable and significant in both subjects. The interaction loading on Emotional Stability is therefore noted but not adopted as the headline. Section 4 then asks whether a more flexible functional form preserves the focal SDQ pattern.

Figure 2 illustrates the same pattern visually. Each panel shows predicted achievement as a function of cognitive ability, evaluated at low ( $-1$  SD), median, and high ( $+1$  SD) Focused Behaviour under the preferred Model 4 specification. The diverging slopes show that the marginal return to cognitive ability is steeper for students with lower Focused Behaviour and shallower for those already strong on the noncognitive margin. Total predicted achievement is still higher for students with stronger attentional regulation at any given level of cognitive ability; the figure shows that an additional unit of cognitive ability translates into more additional achievement when attentional regulation is low.

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<sup>3</sup>This linear cross-effect is distinct from the nonlinear substitution object considered later in the translog sensitivity section.

Table 6: Headline interaction coefficients from the preferred baseline-control linear specification (Model 4), contrasting SDQ Focused Behaviour with TIPI Conscientiousness.

| Interaction                                 | Maths                | English              |
|---------------------------------------------|----------------------|----------------------|
| Cognition $\times$ Focused Behaviour (SDQ)  | −0.052***<br>(0.017) | −0.062***<br>(0.016) |
| Cognition $\times$ Conscientiousness (TIPI) | −0.014<br>(0.015)    | −0.015<br>(0.014)    |

Note: Model 4 adds Wave 1 reading, maths, and SDQ baseline controls to the Model 3 specification. HC1 robust standard errors in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ . Sample sizes are  $N = 3,711$  for the SDQ specifications and  $N = 3,708$  for the TIPI specifications.

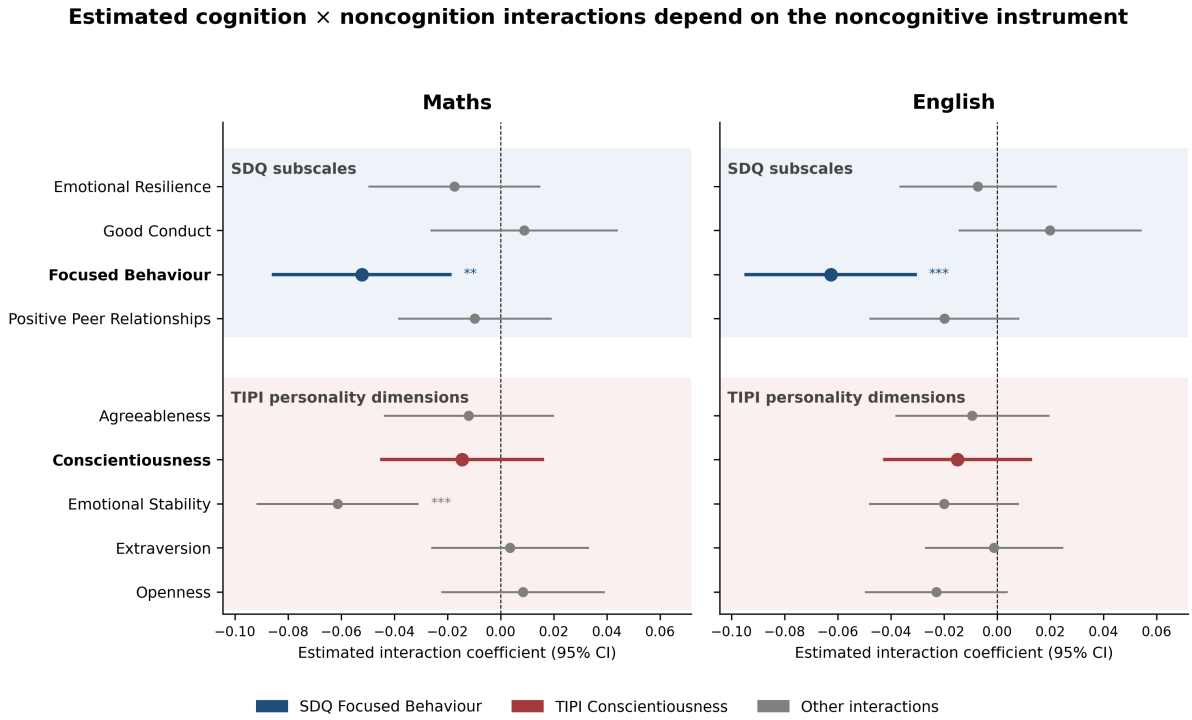


Figure 1: Cognition  $\times$  noncognition interaction by noncognitive instrument. Each point reports the estimated cognition  $\times$  noncognition interaction coefficient from the preferred baseline-control linear specification (Model 4). Horizontal bars are 95% confidence intervals based on HC1 robust standard errors. Significance is reported with the chapter’s standard convention (\* $p < 0.1$ , \*\* $p < 0.05$ , \*\*\* $p < 0.01$ ). The primary noncognitive measure (SDQ Focused Behaviour) is highlighted in blue and the confirmatory measure (TIPI Conscientiousness) in red; the remaining seven interactions from the SDQ and TIPI grids are shown in grey. The SDQ Focused Behaviour interaction is negative and statistically significant in both Maths and English, while the corresponding TIPI Conscientiousness interaction is small and imprecisely estimated in both outcomes. Sample:  $N = 3,711$  for SDQ specifications and  $N = 3,708$  for TIPI specifications.

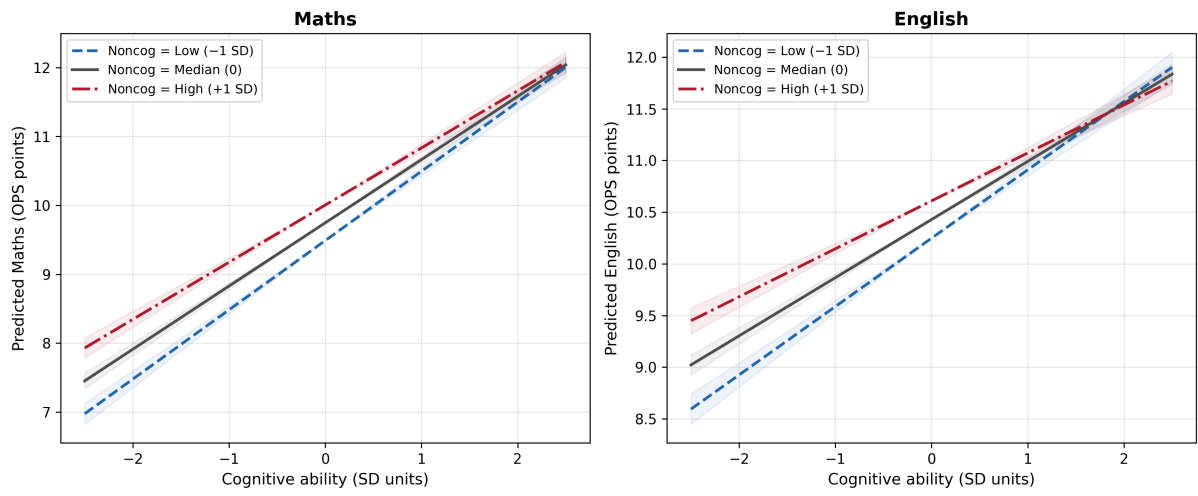


Figure 2: Predicted achievement (OPS points) from the preferred baseline-control linear interaction model (Model 4) with SDQ Focused Behaviour as the noncognitive input, holding all other controls at their sample means. Each line evaluates the marginal effect of cognitive ability (PC1) at a different level of standardised Focused Behaviour ( $-1$  SD, median,  $+1$  SD). The diverging slopes are consistent with the negative cognition  $\times$  Focused Behaviour interaction reported in Table 6. Shaded bands are 95% confidence intervals on the marginal effect, derived from the Model 4 HC1-robust covariance matrix.  $N = 3,711$  for both subjects.



## 4 Nonlinear sensitivity: translog specification

The linear specification establishes that cognitive and noncognitive skills are both associated with achievement and that their interaction terms are predominantly negative. I now ask whether a more flexible functional form preserves that pattern. The translog production function (Christensen et al., 1973) is a second-order Taylor approximation in logarithms that nests the Cobb-Douglas as a special case and so generalises the linear log-log specification with curvature and interaction. I use it here as a flexible local sensitivity check on the linear result, rather than as a literal structural technology - bounded exam scores do not naturally support that interpretation.

### 4.1 Specification

I model achievement as a function of two inputs, cognition ( $C$ ) and a single noncognitive measure ( $N$ ), with SDQ Focused Behaviour as the primary specification and TIPI Conscientiousness as confirmatory. Both specifications are estimated on the same common analytical sample ( $N = 5,631$ ). The translog function is:

$$Y = AC^\alpha N^\beta \exp \left\{ \frac{1}{2} \gamma_1 [\ln(C)]^2 + \frac{1}{2} \gamma_2 [\ln(N)]^2 + \gamma_{12} \ln(C) \ln(N) \right\} \quad (2)$$

where  $Y$  is achievement,  $A$  is a scaling constant,  $\alpha$  and  $\beta$  are the first-order parameters,  $\gamma_1$  and  $\gamma_2$  capture own-curvature, and  $\gamma_{12}$  captures the cross-input interaction - how the marginal product of one input depends on the level of the other. For estimation, I work with the equivalent additive log form and fit it by OLS after centering the log inputs. The local first-order coefficients are therefore read at the centering point in log-input space, and the derived nonlinear summaries reported in the tables below are evaluated at the corresponding sample-mean reference point.

The implied output elasticities vary with input levels:

$$OE_C = \alpha + \gamma_1 \ln(C) + \gamma_{12} \ln(N), \quad OE_N = \beta + \gamma_2 \ln(N) + \gamma_{12} \ln(C).$$

Inference on the derived quantities reported below uses a nonparametric bootstrap with 1,000 replications.

Because gender heterogeneity is a secondary dimension of the chapter, the tables report subgroup estimates descriptively; pooled gender-difference tests are reported in Section 5.

### 4.2 Results

Tables 7 and 8 report the translog estimates for Maths and English. I focus on three quantities: the output elasticities, the cross-curvature parameter  $\gamma_{12}$ , and the SDQ-vs-TIPI

contrast in that parameter, summarised in Table 9.

Table 7: Translog production-function estimates for Junior Certificate Maths (OPS points, 2–12 scale). SDQ Focused Behaviour is the primary noncognitive input; TIPI Conscientiousness is confirmatory. Output elasticities (OEs) give the percentage change in achievement for a 1% change in the input. The marginal rate of technical substitution (MRTS) is reported as a local descriptive summary only and is not used for headline interpretation. The derived quantities are evaluated at the sample-mean reference point after centering the log inputs.  $N = 5,631$  (common analytical sample, identical for SDQ and TIPI by construction).

| Parameter             | Maths             |                   |                   |                |
|-----------------------|-------------------|-------------------|-------------------|----------------|
|                       | Full              | Boys              | Girls             | $\Delta$ (B–G) |
| TIPI Model            |                   |                   |                   |                |
| A                     | 9.43***<br>(0.03) | 9.35***<br>(0.04) | 9.51***<br>(0.04) | –0.16          |
| $\alpha$ (Cognition)  | 0.83***<br>(0.02) | 0.85***<br>(0.02) | 0.82***<br>(0.02) | 0.03           |
| $\beta$ (Consc.)      | 0.04***<br>(0.01) | 0.04***<br>(0.01) | 0.05***<br>(0.01) | –0.01          |
| $\gamma_1$            | 0.10<br>(0.11)    | 0.31*<br>(0.18)   | –0.15<br>(0.16)   | 0.46           |
| $\gamma_2$            | 0.05***<br>(0.01) | 0.03*<br>(0.02)   | 0.07***<br>(0.02) | –0.04          |
| $\gamma_{12}$         | –0.04<br>(0.03)   | –0.09**<br>(0.04) | 0.004<br>(0.04)   | –0.09          |
| MP (Cognition)        | 0.078             | 0.079             | 0.079             | 0.000          |
| MP (Consc.)           | 0.097             | 0.081             | 0.107             | –0.026         |
| OE (Cognition)        | 0.824             | 0.848             | 0.821             | 0.027          |
| OE (Consc.)           | 0.049             | 0.040             | 0.056             | –0.016         |
| MRTS                  | 0.804             | 0.977             | 0.736             | 0.241          |
| SDQ Model             |                   |                   |                   |                |
| A                     | 9.45***<br>(0.02) | 9.35***<br>(0.03) | 9.54***<br>(0.03) | –0.19          |
| $\alpha$ (Cognition)  | 0.79***<br>(0.02) | 0.81***<br>(0.02) | 0.78***<br>(0.02) | 0.03           |
| $\beta$ (Foc. Behav.) | 0.10***<br>(0.01) | 0.09***<br>(0.01) | 0.12***<br>(0.01) | –0.03          |
| $\gamma_1$            | 0.20*<br>(0.11)   | 0.37**<br>(0.18)  | 0.04<br>(0.16)    | 0.33           |
| $\gamma_2$            | 0.06***<br>(0.01) | 0.05***<br>(0.01) | 0.08***<br>(0.02) | –0.03          |
| $\gamma_{12}$         | –0.13***          | –0.11***          | –0.15***          | 0.04           |

*Continued on next page*

Table 7 continued

| Parameter        | Maths  |        |        |                |
|------------------|--------|--------|--------|----------------|
|                  | Full   | Boys   | Girls  | $\Delta$ (B–G) |
|                  | (0.03) | (0.04) | (0.05) |                |
| MP (Cognition)   | 0.074  | 0.075  | 0.074  | 0.001          |
| MP (Foc. Behav.) | 0.128  | 0.109  | 0.146  | –0.037         |
| OE (Cognition)   | 0.777  | 0.803  | 0.770  | 0.033          |
| OE (Foc. Behav.) | 0.109  | 0.088  | 0.129  | –0.041         |
| MRTS             | 0.576  | 0.694  | 0.508  | 0.186          |
| Observations     | 5,631  | 2,667  | 2,801  |                |

*SEs in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .  $\Delta$  is descriptive.*

Table 8: Translog production-function estimates for Junior Certificate English (OPS points, 5–12 scale). SDQ Focused Behaviour is the primary noncognitive input; TIPI Conscientiousness is confirmatory. Output elasticities (OEs) give the percentage change in achievement for a 1% change in the input. MRTS is reported as a local descriptive summary only and is not used for headline interpretation. The derived quantities are evaluated at the sample-mean reference point after centering the log inputs.  $N = 5,631$  (common analytical sample, identical for SDQ and TIPI by construction).

| Parameter            | English            |                   |                    |                |
|----------------------|--------------------|-------------------|--------------------|----------------|
|                      | Full               | Boys              | Girls              | $\Delta$ (B–G) |
| TIPI Model           |                    |                   |                    |                |
| A                    | 10.09***<br>(0.02) | 9.84***<br>(0.04) | 10.36***<br>(0.03) | –0.52          |
| $\alpha$ (Cognition) | 0.45***<br>(0.01)  | 0.51***<br>(0.02) | 0.44***<br>(0.02)  | 0.07           |
| $\beta$ (Consc.)     | 0.02***<br>(0.00)  | 0.02***<br>(0.01) | 0.02***<br>(0.01)  | 0.00           |
| $\gamma_1$           | –0.25***<br>(0.09) | –0.25<br>(0.15)   | –0.32***<br>(0.12) | 0.07           |
| $\gamma_2$           | 0.03***<br>(0.01)  | 0.03*<br>(0.02)   | 0.02<br>(0.01)     | 0.01           |
| $\gamma_{12}$        | –0.03<br>(0.02)    | –0.04<br>(0.03)   | –0.01<br>(0.03)    | –0.03          |
| MP (Cognition)       | 0.046              | 0.049             | 0.047              | 0.002          |
| MP (Consc.)          | 0.059              | 0.048             | 0.039              | 0.009          |
| OE (Cognition)       | 0.450              | 0.501             | 0.446              | 0.055          |
| OE (Consc.)          | 0.028              | 0.022             | 0.019              | 0.003          |
| MRTS                 | 0.775              | 1.027             | 1.207              | –0.180         |

*Continued on next page*

Table 8 continued

| Parameter             | English            |                    |                    |                |
|-----------------------|--------------------|--------------------|--------------------|----------------|
|                       | Full               | Boys               | Girls              | $\Delta$ (B–G) |
| SDQ Model             |                    |                    |                    |                |
| A                     | 10.08***<br>(0.02) | 9.83***<br>(0.03)  | 10.34***<br>(0.03) | –0.51          |
| $\alpha$ (Cognition)  | 0.41***<br>(0.01)  | 0.47***<br>(0.02)  | 0.41***<br>(0.02)  | 0.06           |
| $\beta$ (Foc. Behav.) | 0.09***<br>(0.01)  | 0.07***<br>(0.01)  | 0.08***<br>(0.01)  | –0.01          |
| $\gamma_1$            | –0.17*<br>(0.09)   | –0.14<br>(0.15)    | –0.17<br>(0.12)    | 0.03           |
| $\gamma_2$            | 0.06***<br>(0.01)  | 0.05***<br>(0.01)  | 0.06***<br>(0.01)  | –0.01          |
| $\gamma_{12}$         | –0.12***<br>(0.02) | –0.12***<br>(0.03) | –0.15***<br>(0.04) | 0.03           |
| MP (Cognition)        | 0.041              | 0.045              | 0.043              | 0.002          |
| MP (Foc. Behav.)      | 0.116              | 0.094              | 0.100              | –0.006         |
| OE (Cognition)        | 0.406              | 0.456              | 0.409              | 0.047          |
| OE (Foc. Behav.)      | 0.092              | 0.073              | 0.081              | –0.008         |
| MRTS                  | 0.356              | 0.476              | 0.430              | 0.046          |
| Observations          | 5,631              | 2,667              | 2,801              |                |

*SEs in parentheses. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .  $\Delta$  is descriptive.*

**Output elasticities.** The cognitive output elasticities are large and significant in both subjects ( $OE_C \approx 0.78$ – $0.82$  for Maths,  $0.41$ – $0.45$  for English). The attentional-regulation elasticities are smaller but consistently positive, and the construct-sensitivity gap visible in the linear section reappears here: the SDQ Focused Behaviour measure gives  $OE_N = 0.109$  in Maths and  $0.092$  in English, roughly twice the size of the corresponding TIPI Conscientiousness elasticities ( $0.049$  and  $0.028$ ). At sample means, a 10% increase in cognition is associated with a 7.8% increase in Maths points and a 4.1% increase in English points, while a 10% increase in Focused Behaviour is associated with a 1.1% increase in Maths and 0.9% in English - smaller in magnitude but still positive.

**Cross-curvature.** The interaction parameter  $\gamma_{12}$  is negative and significant in the SDQ models for both subjects, at approximately  $-0.13$  for Maths and  $-0.12$  for English (Table 9). A negative  $\gamma_{12}$  implies that the marginal product of each input decreases as the other rises, which is consistent with the same negative cross-effect pattern seen in the linear interaction models. Because the translog is estimated in log-log form,  $\gamma_{12}$  is not directly comparable in magnitude to the linear interaction coefficient; the substantively

reliable result from the translog is the sign and its persistence across specifications.

**Construct sensitivity.** The SDQ-vs-TIPI contrast carries through to the translog. The TIPI specifications give  $\gamma_{12}$  estimates of  $-0.04$  in Maths and  $-0.03$  in English, neither significant, whereas the SDQ specifications give  $\gamma_{12} \approx -0.12$  to  $-0.13$  and significant in both subjects. The translog therefore reproduces the linear-section finding that the recovered cross-effect depends on which noncognitive construct is used.

Table 9: Headline cross-curvature estimates ( $\gamma_{12}$ ) from the translog specifications, contrasting SDQ Focused Behaviour with TIPI Conscientiousness. Estimates are from the full-sample translog and parallel the linear-section headline in Table 6.

| $\gamma_{12}$          | Maths                   | English                 |
|------------------------|-------------------------|-------------------------|
| SDQ Focused Behaviour  | $-0.13^{***}$<br>(0.03) | $-0.12^{***}$<br>(0.02) |
| TIPI Conscientiousness | $-0.04$<br>(0.03)       | $-0.03$<br>(0.02)       |

Note: Full-sample translog estimates ( $N = 5,631$ , common analytical sample). Bootstrap standard errors in parentheses (1,000 replications).  $***p < 0.01$ ,  $**p < 0.05$ ,  $*p < 0.1$ .

**Gender heterogeneity.** Gender differences in the core parameters are present but moderate: boys show slightly larger cognitive elasticities and girls slightly larger attentional-regulation elasticities, but the broad ranking of inputs is unchanged. The negative cross-curvature pattern is present in both genders and is slightly stronger among girls in the SDQ Maths specification, though the pooled gender-difference tests reported in Section 5 do not reject equality.

**Summary.** The output elasticities and cross-curvature signs are the quantities I rely on from the translog exercise. The MRTS entries are retained as local descriptive summaries but do not support a substitution-rate claim. The local elasticity formula and its interpretation are summarised in Appendix 6.7.

## 5 Inference and sensitivity

This section reports the inference and robustness work for the linear and translog specifications. Linear-model inference uses HC1 standard errors throughout; translog inference uses 1,000-replication nonparametric bootstrap with 95% percentile intervals. Table 10 summarises the linear-model robustness checks: the focal cognition  $\times$  Focused Behaviour interaction is negative in every cell of Panel A, and no check reverses the sign. The rest of the section discusses each check in turn, with translog-specific concerns and identification-style sensitivities at the end.

**Wave 1 baseline controls.** In the preferred baseline-control specification,<sup>4</sup> I re-estimate the interaction models with Wave 1 (age 9) controls included: standardised Wave 1 reading and maths test scores and the four parent-reported SDQ subscales at age 9. The aim is to absorb stable child-level traits and early-environment factors that might otherwise confound the association between Wave 2 inputs and Junior Certificate outcomes. Wave 1 baselines are individually significant and increase  $R^2$  by 3–6 percentage points. All 18 cognition $\times$ noncognition interaction terms preserve their signs after the addition of Wave 1 controls, with a median absolute attenuation of about 20%. The focal SDQ Focused Behaviour  $\times$  Cognition interaction moves from  $-0.056$  to  $-0.052$  in Maths and from  $-0.072$  to  $-0.062$  in English. The interaction pattern is therefore not simply picking up time-invariant ability or temperament (Table 11). These comparisons are estimated on the Model 4 sample to hold the observation set constant, so the Model 3 coefficients here differ slightly from the larger-sample Model 3 coefficients in Table 4.

**Cross-informant robustness of Focused Behaviour.** A natural concern with the SDQ-based result is that parent-rated noncognitive scores partly reflect parental reporting style. If parents who report more child problems also differ on correlated unobservables, the focal cognition  $\times$  Focused Behaviour interaction could pick up reporting bias rather than the underlying attentional-regulation margin. The GUI design allows this concern to be addressed directly: the Wave 1 questionnaire collected the same SDQ subscales from both the primary caregiver and the child’s class teacher.

I re-estimate the preferred baseline-control specification using three variants of Wave 1 SDQ Focused Behaviour: parent-rated (PCG), teacher-rated (TC), and a simple multi-informant composite formed as the unweighted average of the standardised parent and teacher scores. The cognitive input remains the Wave 2 PC composite used in the main

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<sup>4</sup>I use “baseline control” instead of “value-added” to describe the inclusion of Wave 1 skill measures, following the taxonomy of Todd and Wolpin (2003): a proper value-added specification requires lagged *same* outcome scores, whereas the present design conditions on earlier-wave skill inputs that are related to but not identical to Junior Certificate grades. The distinction affects the interpretation of omitted-variable bias.

Table 10: Linear-model robustness summary for the primary SDQ Focused Behaviour  $\times$  Cognition interaction. Panel A reports the interaction coefficient under progressively more demanding specifications, including a cross-informant variant in which the noncognitive input is rated by the child’s class teacher rather than the parent. Panel B reports inference under conservative clustering assumptions and multiple-testing adjustment. The interaction is negative in every cell of Panel A; no linear-model check reverses the sign.

|                                                | Specification                                        | Maths                               | English                             |
|------------------------------------------------|------------------------------------------------------|-------------------------------------|-------------------------------------|
| <i>Panel A: Linear interaction coefficient</i> |                                                      |                                     |                                     |
| (1)                                            | OLS with controls (Model 3)                          | −0.057***                           | −0.074***                           |
| (2)                                            | + Baseline controls (Model 4, preferred)             | −0.052***                           | −0.062***                           |
| (3)                                            | IPW-reweighted (selection correction)                | −0.074**                            | −0.100***                           |
| (4)                                            | Z-score outcome (second-order polynomial)            | −0.036***                           | −0.056***                           |
| (5)                                            | Cognition leave-one-out range                        | [−0.046, −0.061]<br>all $p < 0.05$  | [−0.061, −0.069]<br>all $p < 0.001$ |
| (6)                                            | Cross-informant: teacher-rated W1 SDQ (Model 4)      | −0.065**                            | −0.069***                           |
| (7)                                            | Cross-informant: combined PCG+TC composite (Model 4) | −0.074***                           | −0.090***                           |
| <i>Panel B: Inference sensitivity</i>          |                                                      |                                     |                                     |
| (8)                                            | Design-effect ( $\rho = 0.20$ , DEFF = 2.40)         | $t = -1.99$<br>( $p \approx 0.05$ ) | $t = -2.49$<br>( $p < 0.05$ )       |
| (9)                                            | BH-FDR $q$ -value (18 interaction tests)             | $q = 0.012$                         | $q = 0.001$                         |

*Notes:* Panel A reports Cognition  $\times$  SDQ Focused Behaviour interaction coefficients. Row (1): OLS with SES and school controls ( $N \approx 3,786$ ). Row (2): adds Wave 1 age-9 baseline reading, maths, and SDQ subscales. Row (3): inverse-probability-weighted for selection into the restricted sample, applied to the uncontrolled specification (baseline unweighted coefficients are −0.083 for Maths and −0.098 for English, so the IPW modestly attenuates the interaction in Maths and is approximately neutral in English). Row (4): dependent variable standardised to  $z$ -scores; cross-term from second-order polynomial. Row (5): cognition reconstructed by dropping each of three subtests in turn. Rows (6)–(7): Wave 2 cognition paired with a Wave 1 SDQ Focused Behaviour input rated by the class teacher (row 6) or by an unweighted parent–teacher composite (row 7), under the preferred baseline-control specification (Model 4). See Table 12 for the full cross-informant table. Panel B: Row (8) inflates HC1 SEs by  $\sqrt{\text{DEFF}}$  under the most conservative intra-class correlation considered ( $\rho = 0.20$ , average cluster size  $\bar{m} = 8$ ). Row (9): Benjamini-Hochberg adjusted  $q$ -values across 18 interaction tests. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

Table 11: OLS interaction coefficients with and without Wave 1 baseline controls (preferred baseline-control specification).

| Spec                                                    | Interaction     | Model 4 (+W1, preferred) |         | Model 3   |         | Change<br>(%) |
|---------------------------------------------------------|-----------------|--------------------------|---------|-----------|---------|---------------|
|                                                         |                 | Coef                     | SE      | Coef      | SE      |               |
| <i>Panel A: SDQ (Focused Behaviour) - Primary</i>       |                 |                          |         |           |         |               |
| Maths                                                   | Emot. Res.×Cog  | −0.017                   | (0.016) | −0.028*   | (0.016) | +38           |
| Maths                                                   | Good Cond.×Cog  | 0.009                    | (0.018) | 0.012     | (0.018) | −26           |
| Maths                                                   | Foc. Behav.×Cog | −0.052***                | (0.017) | −0.056*** | (0.017) | +7            |
| Maths                                                   | Pos. Peers×Cog  | −0.010                   | (0.015) | −0.016    | (0.014) | +38           |
| English                                                 | Emot. Res.×Cog  | −0.007                   | (0.015) | −0.015    | (0.015) | +51           |
| English                                                 | Good Cond.×Cog  | 0.020                    | (0.017) | 0.024     | (0.018) | −19           |
| English                                                 | Foc. Behav.×Cog | −0.062***                | (0.016) | −0.072*** | (0.017) | +14           |
| English                                                 | Pos. Peers×Cog  | −0.020                   | (0.014) | −0.026*   | (0.015) | +23           |
| <i>Panel B: TIPI (Conscientiousness) - Confirmatory</i> |                 |                          |         |           |         |               |
| Maths                                                   | Agreeable×Cog   | −0.012                   | (0.016) | −0.017    | (0.017) | +28           |
| Maths                                                   | Conscient.×Cog  | −0.014                   | (0.015) | −0.017    | (0.016) | +13           |
| Maths                                                   | Emo. Stab.×Cog  | −0.061***                | (0.015) | −0.067*** | (0.016) | +8            |
| Maths                                                   | Extravert×Cog   | 0.004                    | (0.015) | 0.001     | (0.015) | -             |
| Maths                                                   | Openness×Cog    | 0.008                    | (0.016) | 0.007     | (0.016) | +14           |
| English                                                 | Agreeable×Cog   | −0.009                   | (0.015) | −0.012    | (0.015) | +20           |
| English                                                 | Conscient.×Cog  | −0.015                   | (0.014) | −0.017    | (0.015) | +15           |
| English                                                 | Emo. Stab.×Cog  | −0.020                   | (0.014) | −0.026*   | (0.015) | +23           |
| English                                                 | Extravert×Cog   | −0.001                   | (0.013) | −0.006    | (0.014) | +80           |
| English                                                 | Openness×Cog    | −0.023*                  | (0.014) | −0.024*   | (0.014) | +3            |

Note: Model 4 (preferred) includes Wave 2 cognitive ability (PC), noncognitive dimensions, SES and school controls, cognition  $\times$  noncognition interaction terms, and standardised Wave 1 baselines (reading score, maths score, and four parent-reported SDQ subscales at age 9). Model 3 omits Wave 1 baselines. All 18 interaction terms preserve their signs. Change (%) is the percentage change in the absolute size of the coefficient between Model 3 and Model 4; positive values indicate attenuation toward zero. HC1 robust standard errors. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .



specification, so the only change relative to the baseline-control model in Table 11 is the source of the noncognitive measure. The multi-informant composite is treated descriptively, as an unweighted average of the two informants' standardised scores, rather than as a latent-factor or errors-in-variables correction.

Cross-informant agreement is highest for the hyperactivity subscale ( $r \approx 0.43$ ), followed by peer problems ( $r \approx 0.29$ ), conduct problems ( $r \approx 0.24$ ), and emotional symptoms ( $r \approx 0.21$ ). The relative ordering is consistent with the broader psychometric literature: externalising and attentional behaviours are more visible across raters than internalising or relational difficulties. This pattern helps explain why Focused Behaviour is the most empirically stable SDQ dimension throughout.

Table 12 reports the cognition  $\times$  Focused Behaviour interaction across the three informant variants, with and without Wave 1 baseline controls. All twelve cells are negative and statistically significant at the 5% level or better. Magnitudes are broadly similar across informants for both outcomes, with somewhat more variation in English than in Maths: in Maths, the parent-rated interaction is  $-0.066$ , the teacher-rated interaction is  $-0.065$ , and the combined composite is  $-0.074$  in the preferred baseline-control specification; in English, the corresponding values are  $-0.087$ ,  $-0.069$ , and  $-0.090$ . The interaction sign and approximate magnitude are therefore not specific to any single informant.

The result strengthens the central claim in two ways. First, the negative cognition  $\times$  Focused Behaviour interaction is not an artefact of parent-specific reporting style: a teacher-rated version of the same construct, collected at age 9, gives the same qualitative pattern when used in the main specification. Second, the result holds when the noncognitive input is moved four years earlier in the developmental window, which makes a story based on contemporaneous parent-child reporting interactions less plausible. What survives across informants and across measurement timing is the negative interaction attached to attentional regulation.

**Functional form: z-score outcome.** The main interaction result does not depend on the log transformation. A second-order polynomial with z-scored outcomes produces the same sign pattern across all 12 specifications (Appendix 6.9, Table 18).

**Construct-difference test.** The chapter's central methodological claim - that the recovered interaction depends on which noncognitive construct is used - can be tested directly. A same-sample test rejects equality between the SDQ Focused Behaviour  $\times$  Cognition and TIPI Conscientiousness  $\times$  Cognition interaction coefficients in both subjects, both with and without Wave 1 baseline controls (Appendix 6.9, Table 19). The construct-sensitivity finding is therefore not just a difference in significance under separate models; the two interaction coefficients are statistically distinguishable from one another.

Table 12: Cross-informant robustness of the cognition  $\times$  Focused Behaviour interaction. Each row uses the main Wave 2 cognitive composite and a Wave 1 SDQ Hyperactivity (Focused Behaviour) measure from one of three informants. Model 3 includes SES and school controls; Model 4 (preferred) adds Wave 1 baseline controls. HC1 robust standard errors.

| Informant    | Model   | Maths     |         | English   |         |
|--------------|---------|-----------|---------|-----------|---------|
|              |         | Coef      | SE      | Coef      | SE      |
| Parent (PCG) | Model 3 | −0.075*** | (0.019) | −0.101*** | (0.019) |
| Parent (PCG) | Model 4 | −0.066*** | (0.019) | −0.087*** | (0.018) |
| Teacher (TC) | Model 3 | −0.069*** | (0.021) | −0.082*** | (0.021) |
| Teacher (TC) | Model 4 | −0.065*** | (0.022) | −0.069*** | (0.020) |
| Combined     | Model 3 | −0.085*** | (0.019) | −0.108*** | (0.018) |
| Combined     | Model 4 | −0.074*** | (0.020) | −0.090*** | (0.018) |

*Notes:* Each cell reports the Cognition  $\times$  Focused Behaviour interaction coefficient. Cognition is the Wave 2 PC composite (Drumcondra Verbal Reasoning, Drumcondra Numerical Ability, BAS Matrix Reasoning). Focused Behaviour is the Wave 1 SDQ Hyperactivity subscale rated by the parent (PCG), the class teacher (TC), or a simple unweighted average of the two standardised scores (Combined). Model 3 conditions on SES and school controls; Model 4 adds Wave 1 baselines (Wave 1 cognitive composites and the four Wave 1 PCG SDQ subscales, excluding the focal Hyperactivity subscale where it would collide with the focal informant).  $N$  ranges from 3,702 to 3,782. \*\*\* $p < 0.01$ , \*\* $p < 0.05$ , \* $p < 0.1$ .

**Multiple-testing correction.** The analysis estimates multiple specifications across instruments, subjects, and subgroups, so I follow a pre-specified hierarchy: SDQ Focused Behaviour is primary, TIPI Conscientiousness is confirmatory, and broader grids are exploratory. To keep that transparent, I report Benjamini–Hochberg false discovery rate (FDR)  $q$ -values for the 18 cognition  $\times$  noncognition interaction tests in the baseline-control specifications (Table 13). Three results survive the 10% FDR threshold: the two primary SDQ Focused Behaviour  $\times$  Cognition interactions in Maths and English, and TIPI Emotional Stability  $\times$  Cognition in Maths.

**The Emotional Stability survivor.** The TIPI Emotional Stability survivor is worth addressing, but it does not overturn the chapter’s main measurement contrast. The pre-specified TIPI comparator is Conscientiousness, and that focal interaction remains small and imprecise throughout. The Emotional Stability result is best read as evidence that the broader behavioural-regulation margin may extend beyond Conscientiousness within the TIPI grid, not as a refutation of the finding that the clearest and most stable interaction loads on the school-proximal SDQ Focused Behaviour measure.

**Pooled gender-difference tests.** To complement the subgroup analyses, I also estimate pooled baseline-control models with full gender interactions and joint Wald tests (Table 14). The null that all gender differences in cognition  $\times$  noncognition interactions are zero is not rejected in the four pooled specifications ( $p$ -values 0.29–0.65). The subgroup contrasts

Table 13: Interaction-grid p-values with Benjamini-Hochberg false-discovery-rate (FDR) adjustment. The analysis tests 18 separate cognition  $\times$  noncognitive interactions ((4 SDQ subscales + 5 TIPI dimensions)  $\times$  2 subjects), so some may appear significant by chance alone. The  $q_{BH}$  column adjusts for this multiple testing: it controls the expected share of false positives among all rejected hypotheses. Baseline-control models (Wave 1 baselines included).

| Outcome | Spec | Interaction                       | Coef    | SE     | p      | $q_{BH}$            |
|---------|------|-----------------------------------|---------|--------|--------|---------------------|
| Maths   | SDQ  | SDQ Emot. Res. $\times$ Cog       | -0.0172 | 0.0162 | 0.2870 | 0.5740              |
| Maths   | SDQ  | SDQ Good Cond. $\times$ Cog       | 0.0088  | 0.0176 | 0.6169 | 0.7046              |
| Maths   | SDQ  | SDQ Foc. Behav. $\times$ Cog      | -0.0517 | 0.0168 | 0.0021 | 0.0124 <sup>†</sup> |
| Maths   | SDQ  | SDQ Pos. Peers $\times$ Cog       | -0.0097 | 0.0145 | 0.5034 | 0.7046              |
| Maths   | TIPI | Agreeable W2 PCG $\times$ Cog     | -0.0119 | 0.0162 | 0.4608 | 0.7046              |
| Maths   | TIPI | Conscientious W2 PCG $\times$ Cog | -0.0144 | 0.0154 | 0.3489 | 0.6280              |
| Maths   | TIPI | Emo Stability W2 PCG $\times$ Cog | -0.0614 | 0.0154 | 0.0001 | 0.0010 <sup>†</sup> |
| Maths   | TIPI | Extravert W2 PCG $\times$ Cog     | 0.0036  | 0.0150 | 0.8094 | 0.8570              |
| Maths   | TIPI | Openness W2 PCG $\times$ Cog      | 0.0084  | 0.0155 | 0.5863 | 0.7046              |
| English | SDQ  | SDQ Emot. Res. $\times$ Cog       | -0.0072 | 0.0147 | 0.6263 | 0.7046              |
| English | SDQ  | SDQ Good Cond. $\times$ Cog       | 0.0196  | 0.0171 | 0.2528 | 0.5740              |
| English | SDQ  | SDQ Foc. Behav. $\times$ Cog      | -0.0620 | 0.0161 | 0.0001 | 0.0010 <sup>†</sup> |
| English | SDQ  | SDQ Pos. Peers $\times$ Cog       | -0.0198 | 0.0141 | 0.1612 | 0.4836              |
| English | TIPI | Agreeable W2 PCG $\times$ Cog     | -0.0094 | 0.0146 | 0.5193 | 0.7046              |
| English | TIPI | Conscientious W2 PCG $\times$ Cog | -0.0149 | 0.0140 | 0.2863 | 0.5740              |
| English | TIPI | Emo Stability W2 PCG $\times$ Cog | -0.0201 | 0.0142 | 0.1587 | 0.4836              |
| English | TIPI | Extravert W2 PCG $\times$ Cog     | -0.0013 | 0.0131 | 0.9227 | 0.9227              |
| English | TIPI | Openness W2 PCG $\times$ Cog      | -0.0230 | 0.0135 | 0.0884 | 0.3978              |

Note: Rows report cognition  $\times$  noncognition interactions from baseline-control OLS models (Wave 1 baselines included).  $q$ -values are Benjamini-Hochberg adjusted across all 18 interaction tests. <sup>†</sup> indicates  $q_{BH} < 0.10$ .

therefore suggest heterogeneity, but gender differences in the interaction structure are not detectable at standard levels. A power calculation contextualises this null: with  $N \approx 1,850$  per gender subgroup and a pooled interaction SE of approximately 0.017 for the focal SDQ term, the minimum detectable effect (MDE) for a gender difference in the interaction coefficient is approximately 0.10 at 80% power ( $\alpha = 0.05$ ).<sup>5</sup> Since the main interaction is only  $-0.052$  in Maths, the test detects gender differences only if they are roughly twice the size of the main effect itself - a demanding threshold, and consistent with modest or absent gender heterogeneity.

Table 14: Pooled gender-difference joint tests in baseline-control interaction models.

| Outcome | Spec | N    | p(Joint gender $\times$ interaction = 0) | p(Joint all gender-diff terms = 0) |
|---------|------|------|------------------------------------------|------------------------------------|
| Maths   | SDQ  | 3711 | 0.3525                                   | 0.3087                             |
| Maths   | TIPI | 3708 | 0.4430                                   | 0.7152                             |
| English | SDQ  | 3711 | 0.6531                                   | 0.6732                             |
| English | TIPI | 3708 | 0.2935                                   | 0.6727                             |

Note: HC1-robust Wald tests from pooled models with Wave 1 baseline controls. The first p-value tests whether all gender differences in cognition $\times$ noncognition interactions are zero; the second tests all gender-difference terms jointly.

**Cognition stress test.** I stress-test the cognition construct by rebuilding PC under leave-one-subtest-out variants and re-estimating the primary SDQ baseline-control interaction model (Table 15). The focal Focused Behaviour  $\times$  Cognition interaction remains negative in all variants for both outcomes, with statistical significance preserved at conventional levels (Maths p-values 0.001–0.035; English p-values below 0.001). The main interaction pattern is therefore not an artefact of any single cognitive subtest.

Table 15: Cognition stress test: leave-one-subtest-out PC and key interaction stability.

| Outcome | Cognition variant                | N    | $\hat{\beta}_{FB \times Cog}$ | SE     | p      |
|---------|----------------------------------|------|-------------------------------|--------|--------|
| Maths   | All three subtests (baseline PC) | 3711 | $-0.0517$                     | 0.0168 | 0.0021 |
| English | All three subtests (baseline PC) | 3711 | $-0.0620$                     | 0.0161 | 0.0001 |
| Maths   | Drop Drum_VR_W2_p                | 3711 | $-0.0513$                     | 0.0208 | 0.0137 |
| English | Drop Drum_VR_W2_p                | 3711 | $-0.0614$                     | 0.0178 | 0.0006 |
| Maths   | Drop Drum_NA_W2_p                | 3711 | $-0.0462$                     | 0.0219 | 0.0350 |
| English | Drop Drum_NA_W2_p                | 3711 | $-0.0685$                     | 0.0192 | 0.0004 |
| Maths   | Drop BAS_TS_Mat_W2               | 3711 | $-0.0605$                     | 0.0183 | 0.0010 |
| English | Drop BAS_TS_Mat_W2               | 3711 | $-0.0668$                     | 0.0179 | 0.0002 |

Note: Primary SDQ baseline-control specification. Each row rebuilds cognition from a different test subset and reports the Focused Behaviour  $\times$  Cognition interaction coefficient.

**Leaving Certificate extension.** As a descriptive extension to a longer prediction horizon, I also explored Leaving Certificate points as the outcome, spanning 4–5 years from Wave 2 inputs to the terminal upper-secondary exam. The first-order pattern remains broadly sensible at that longer horizon, but the nonlinear summaries are still too fragile

<sup>5</sup>Using the approximation  $MDE \approx 2.8 \times SE_{diff}$ , where  $SE_{diff} \approx \sqrt{2} \times SE_{pooled} \times \sqrt{N/N_{sub}} \approx 0.034$ .

to support a separate headline claim. Appendix 6.10 records the scope and limits of that extension.

**Omitted-variable-bias bounds.** Because the design is associational, confounding by unobserved factors remains possible. Oster (2019) bounds for the primary SDQ Focused Behaviour  $\times$  Cognition interaction do not weaken the main result. For English, the bound implies that unobservables would need to be substantially more correlated with the interaction than the included controls to eliminate the effect ( $\delta^* = 4.84$ ). For Maths, adding controls moves the coefficient further from zero rather than toward it, so the standard Oster calculation yields a negative  $\delta^*$ . Under the Oster approximation, this means that any unobservables capable of eliminating the result would have to be correlated with the interaction in the opposite direction from the observed controls - a configuration that is not naturally suggested by the data and that strengthens rather than weakens the interpretation. Full values and sensitivity to  $R_{\max}^2$  are reported in Appendix 6.9, Tables 22 and 23.

**Clustering and design effects.** Standard errors are computed at the individual level rather than clustered at the school level. The GUI Cohort '98 sample was drawn using a two-stage clustered design (Growing Up in Ireland Study Team, 2010), but school identifiers are available only through the Research Microdata File (RMF) pathway; a prior clearance window expired during this work and the re-application process exceeds the current submission timeline.<sup>6</sup>

To gauge the potential magnitude of clustering, I inflate standard errors by factors consistent with the GUI sampling design - roughly 8 sampled students per school and a range of plausible within-school correlations from  $\rho = 0.05$  to  $\rho = 0.20$ . The cited  $\rho$  range covers the values typically reported for Irish school-level student outcomes. Across this range the primary SDQ Focused Behaviour interaction remains significant at the 5% level for both Maths and English (Appendix 6.9, Tables 24 and 25). At  $\rho = 0.20$  the Maths  $t$ -statistic is  $-1.99$ , so significance for Maths would be lost at higher intra-class correlations (around  $\rho \geq 0.25$ ), while English would remain significant at the 5% level up to  $\rho \approx 0.40$ . The Maths result is therefore robust within the empirically plausible range but would not survive an unusually high intra-class correlation.

**TIPI measurement error and construct specificity.** A natural concern with the SDQ-TIPI contrast is that the weaker TIPI interaction may come from higher measurement error in the broader personality proxy. In that case, the contrast would say less about

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<sup>6</sup>Published GUI studies employ a range of inference strategies depending on data access: multilevel models (McCoy & Byrne, 2024) and survey-weighted GEE (Brennan et al., 2025). Several do not cluster at the school level.

substantive differences in what the two instruments capture. The released Wave 2 PCG TIPI aggregates compound this concern in a specific way: the focal Conscientiousness measure has a sample minimum of 0.5 with non-trivial mass at that value, inconsistent with the documented 1–7 item range and indicative of a missing-item handling step in the source construction. To check whether this is keeping the TIPI interaction weak, I re-estimate the preferred baseline-control TIPI specification under two trim variants on the focal Conscientiousness input: a visible trim that drops only the cases below the documented 1.0 minimum (82 cases, 2.2% of the estimation sample), and a conservative ambiguous trim that drops any value at or below 3.5 (1,559 cases), the highest value that could plausibly result from missing-item handling in the released aggregate.

Under the visible trim, the focal Conscientiousness  $\times$  Cognition coefficient moves slightly more negative for both subjects (Maths  $-0.014 \rightarrow -0.016$ ; English  $-0.015 \rightarrow -0.018$ ), in the direction predicted by classical measurement-error attenuation, but remains insignificant. Under the more aggressive ambiguous trim, the Maths coefficient stays near baseline and the English coefficient flips sign while remaining far from significance. Visibly removing the noisiest cases in the focal TIPI input therefore does not pull the interaction toward significance or toward the SDQ magnitude. Differential measurement error in the GUI W2 PCG TIPI aggregate is unlikely to be the main driver of the SDQ–TIPI gap, which is more naturally read as a difference in what each instrument captures.

Full results are in Appendix 6.9, Table 20. Item-level W2 TIPI variables are not in the analytical extract, so a stronger errors-in-variables or SIMEX correction is not currently feasible from the released data.

## 6 Discussion and conclusion

### 6.1 Discussion

**Main result.** The main result is that cognitive ability and attentional regulation interact negatively in predicting Junior Certificate achievement in both Maths and English. The marginal return to cognitive ability is larger for students with weaker attentional regulation, and this pattern survives baseline conditioning, inverse-probability reweighting, alternative functional forms, multiple-testing correction, design-effect-adjusted inference, and a teacher-rated cross-informant version of the same construct collected four years earlier. When the same question is tested using a broader personality proxy, the interaction becomes weak and imprecise. The SDQ-TIPI contrast therefore does substantive work: it shows that the joint-skill relationship recovered in the production function depends on which noncognitive construct is used to represent noncognitive skill, not only on whether a noncognitive input is included at all.

**Cross-informant evidence.** The cross-informant result sharpens that interpretation. The negative interaction survives when the noncognitive input is rated by the child’s class teacher and when it is measured four years earlier in the developmental window. Across raters, timing windows, and specifications, the focal SDQ Focused Behaviour interaction is the most stable interaction in the chapter. This makes a story based mainly on parent-specific reporting style less persuasive and helps clarify which noncognitive margin is most informative in this setting. The SDQ signal that survives the robustness battery is tied to attentional regulation and classroom-relevant behaviour. It should not be read as an abstract residual “noncognitive skill” effect. This distinction is important because the educational production-function literature often treats “noncognitive skill” as a broad residual category, even though the underlying constructs differ substantially in observability, malleability, and institutional relevance (Almlund et al., 2011; Heckman & Kautz, 2012; Humphries & Kosse, 2017). In the present data, the more stable interaction is tied to the behavioural margin that teachers and schools are most likely to observe directly in day-to-day learning environments.

**Gender heterogeneity.** Gender heterogeneity, by contrast, is modest: in the primary SDQ specification, the negative interaction is broadly similar for boys and girls, and the pooled interaction tests do not reject equality.

**Translog as supplementary check.** The translog sensitivity exercise is useful mainly as a supplementary check on the sign of the interaction. Its first-order objects - output elasticities, marginal products, and cross-curvature signs - line up with the linear interaction

picture. The local substitution parameter, however, is empirically fragile and I do not treat it as an independent structural estimate.

**Synthesis.** Taken together, the findings show that cognitive ability and attentional regulation appear to operate as compensating inputs in academic achievement. The data point away from a cleanly additive relationship. The chapter therefore does not defend a stable nonlinear elasticity estimate or a generic claim about all noncognitive measures. It defends a more specific result about the joint role of cognition and attentional regulation, with measurement choice setting the boundary of that claim (Borghans et al., 2008; Humphries & Kosse, 2017).

## 6.2 Limitations

Three categories of limitation shape interpretation.

**Identification.** The design does not identify causal effects. Although predictors are measured two to three years before the outcome, omitted-variable bias remains possible if unobserved factors are correlated with both skill inputs and Junior Certificate outcomes. A specific concern is that the Wave 2 noncognitive measures (age 13) are themselves partially endogenous to prior school experience: a child who has done well at school by age 13 may exhibit better attentional regulation in part because of that experience, not only as an input to later achievement. Conditioning on Wave 1 baselines absorbs prior *levels* of cognition and behaviour at age 9, but it does not break the Wave 1 → Wave 2 dynamic feedback loop in noncognitive development. The cross-informant exercise partially addresses this concern: the negative interaction reappears when Focused Behaviour is measured at Wave 1 (age 9), before most of the Junior-Certificate-relevant school experience has accumulated. The associational interpretation in the chapter already accommodates this, but the interaction should not be read as a causal claim about two independently exogenous inputs.

The complete-case requirement strengthens comparability across specifications and defines the estimand as conditional associations within a selected longitudinal subsample. Selection diagnostics and IPW re-estimation (Appendix 6.8) confirm that this restriction does not drive the main findings, but it does limit generalisability.

**Measurement.** Both noncognitive instruments have known limitations. The TIPI's brevity (ten items covering five personality dimensions) limits its ability to capture nuanced personality traits, and its breadth means it is less directly tied to classroom behaviour than the SDQ. The SDQ relies on a single informant (the primary caregiver at Wave 2) in the baseline specification and may be subject to reporting bias: caregivers may rate



children with stronger school performance more favourably, and a caregiver’s general outlook may inflate correlations across SDQ subscales.

The analysis uses the GUI AMF, which contains the aggregate SDQ subscales and TIPI trait scores but not the underlying item-level responses available through the restricted-access RMF. I therefore cannot estimate within-sample internal-consistency reliability for the focal measures from the analysis files used here.

The focal SDQ and TIPI measures are bounded and show different top-coding patterns in the AMF analysis file. The most pronounced ceiling mass appears in the teacher-rated Wave 1 SDQ and at the upper bound of Wave 2 TIPI Conscientiousness. Appendix Figures 3 and 4 report the corresponding distributions. The cross-informant robustness exercises help reduce concern that the main interaction is driven only by parent-specific reporting style, but they do not fully separate construct specificity from differential reliability. The SDQ–TIPI contrast therefore has limits: it does not fully separate those two channels.

**Inference.** School identifiers are available only through the restricted-access RMF, so school-clustered standard errors are not implemented in the headline tables. The design-effect sensitivity exercise confirms that the main linear interaction finding is robust to plausible clustering adjustments. It serves as a sensitivity analysis and does not replace school-clustered standard errors. For the translog-derived objects, I rely on bootstrap standard errors and interpret them conservatively. My emphasis is on sign, magnitude, and cross-specification stability, with single-threshold significance playing a smaller role.

Despite these limitations, the key patterns replicate across control sets, functional forms, informant sources, and sensitivity checks. The negative interaction under the SDQ Focused Behaviour measure remains the most stable empirical result.

### 6.3 Implications

For education economics, the main implication is that academic achievement depends on cognitive and behavioural skills that work together. The evidence points away from a model in which the inputs operate independently. In the present data, the return to cognitive ability is steepest among students with weaker attentional regulation, which is consistent with a compensatory relationship between the two margins at this developmental stage. That is the substantive contribution of the chapter.

The SDQ–TIPI contrast has a methodological implication beyond this specific finding. When attentional regulation is measured using a school-proximal behavioural instrument, the negative interaction is stable across the main checks reported here. When the same role is assigned to a brief broad personality proxy, the interaction weakens sharply and becomes imprecise.

The chapter also speaks to a broader measurement issue in applied work on child and adolescent skills. Brief omnibus scales are attractive because they are cheap to field and easy to compare across datasets, but that convenience comes at an interpretive cost when the research question concerns how skills combine. That question is different from asking whether each skill predicts outcomes in isolation. For that purpose, multi-item, domain-specific, and ideally multi-informant measures are likely to be more informative than broad summary proxies (Goodman, 1997; Gosling et al., 2003; Humphries & Kosse, 2017).

The findings point to three directions for further work. First, the compensatory interaction reported here should be tested against alternative outcome models that address the bounded nature of the Junior Certificate scale more directly. Second, richer measurement of school-proximal behavioural regulation would help separate construct specificity from differential reliability more cleanly. Third, future work with linked administrative data and restricted-access measurement files could sharpen both inference and measurement.

## 6.4 Conclusion

Within these limits, the chapter provides a descriptive account of how cognition and attentional regulation appear to combine in adolescent academic achievement. The stable empirical result is a negative interaction between cognitive ability and attentional regulation, and the SDQ–TIPI contrast shows that this interaction depends on the behavioural construct used to represent noncognitive skill.

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# Appendices

The following appendices provide the psychometric instruments used in the chapter and the supporting empirical diagnostics referenced in the main text. Appendices A and B reproduce the two psychometric instruments. Appendices C through F document the technical formulas and empirical diagnostics that support the core findings.

1. TIPI questionnaire.
2. SDQ questionnaire.
3. Translog elasticity-of-substitution formula and interpretation.
4. Selection and attrition diagnostics.
5. Additional robustness checks.
6. Complementarity across developmental windows.

## 6.5 TIPI questionnaire

The TIPI (Gosling et al., 2003) measures the Big Five personality traits - Extraversion, Agreeableness, Conscientiousness, Emotional Stability, and Openness to Experience - using ten items (two per trait) on a 7-point Likert scale. Its brevity makes it practical for large-scale surveys such as GUI, though reliability and validity are lower than for full-length inventories. In this study, primary caregivers rated the study child; scoring involves reverse-coding one item per trait and averaging the pair.

|                      |                        |                      |                               |                   |                     |                   |
|----------------------|------------------------|----------------------|-------------------------------|-------------------|---------------------|-------------------|
| Disagree<br>strongly | Disagree<br>moderately | Disagree<br>a little | Neither disagree<br>nor agree | Agree<br>a little | Agree<br>moderately | Agree<br>strongly |
| 1                    | 2                      | 3                    | 4                             | 5                 | 6                   | 7                 |

| Characteristic                      | Rating (1-7) |
|-------------------------------------|--------------|
| 1. Extraverted, enthusiastic        | _____        |
| 2. Critical, quarrelsome            | _____        |
| 3. Dependable, self-disciplined     | _____        |
| 4. Anxious, easily upset            | _____        |
| 5. Open to new experiences, complex | _____        |
| 6. Reserved, quiet                  | _____        |
| 7. Sympathetic, warm                | _____        |
| 8. Disorganized, careless           | _____        |
| 9. Calm, emotionally stable         | _____        |
| 10. Conventional, uncreative        | _____        |

|                                                                                                                                                               |                                                                                                                                                       |                                                                                                                                                                   |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conscientiousness<br>Score for #3: _____<br>+ (8 - Score for #8): _____<br>_____<br>= _____<br>Divide your answer by 2.<br>Conscientiousness Score<br>= _____ | Agreeableness<br>Score for #7: _____<br>+ (8 - Score for #2): _____<br>_____<br>= _____<br>Divide your answer by 2.<br>Agreeableness Score =<br>_____ | Emotional Stability<br>Score for #9: _____<br>+ (8 - Score for #4): _____<br>_____<br>= _____<br>Divide your answer by 2.<br>Emotional Stability Score<br>= _____ |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|                                                                                                                                                                 |                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Openness to Experience<br>Score for #5: _____<br>+ (8 - Score for #10): _____<br>= _____<br>Divide your answer by 2.<br>Openness to Experience Score =<br>_____ | Extraversion<br>Score for #1: _____<br>+ (8 - Score for #6): _____<br>= _____<br>Divide your answer by 2.<br>Extraversion Score = _____ |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|

## 6.6 SDQ questionnaire

The Strengths and Difficulties Questionnaire (Goodman, 1997) is a 25-item behavioural screening tool for children aged 3–16, comprising five subscales: Emotional Symptoms, Conduct Problems, Hyperactivity/Inattention, Peer-relationship Problems, and Prosocial Behaviour. Items are scored on a 3-point scale (“Not True”, “Somewhat True”, “Certainly True”). In this study, the primary caregiver completed the parent version; the Prosocial subscale was excluded because its coding in the anonymised microdata files was not consistent enough across waves to support the same cross-wave treatment as the four difficulty subscales. Higher difficulty subscale scores indicate greater problems (range 0–10 per subscale).

Sample items (from Youth in Mind (2023), <https://sdqinfo.org/>):

### Strengths and Difficulties Questionnaire

For each item, please mark the box for Not True, Somewhat True or Certainly True. Answer all items as best you can even if you are not absolutely certain. Please give your answers on the basis of the child’s behaviour over the last six months.

|                                                                         | Not<br>True              | Somewhat<br>True         | Certainly<br>True        |
|-------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|
| 1. Considerate of other people's feelings                               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 2. Restless, overactive, cannot stay still for long                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 3. Often complains of headaches, stomach-aches or sickness              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 4. Shares readily with other children (treats, toys, pencils etc.)      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 5. Often has temper tantrums or hot tempers                             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 6. Rather solitary, tends to play alone                                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 7. Generally obedient, usually does what adults request                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 8. Many worries, often seems worried                                    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 9. Helpful if someone is hurt, upset or feeling ill                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 10. Constantly fidgeting or squirming                                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 11. Has at least one good friend                                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 12. Often fights with other children or bullies them                    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 13. Often unhappy, down-hearted or tearful                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 14. Generally liked by other children                                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 15. Easily distracted, concentration wanders                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 16. Nervous or clingy in new situations, easily loses confidence        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 17. Kind to younger children                                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 18. Often lies or cheats                                                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 19. Picked on or bullied by other children                              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 20. Often volunteers to help others (parents, teachers, other children) | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 21. Thinks things out before acting                                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 22. Steals from home, school or elsewhere                               | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 23. Gets on better with adults than with other children                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 24. Many fears, easily scared                                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| 25. Sees tasks through to the end, good attention span                  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

The SDQ is divided into five sections, each containing five questions. The Prosocial Scale is shown for documentary completeness, but it is not used in the chapter's analysis:

1. Emotional Symptoms Scale:

- 3. Often complains of headaches, stomach-aches or sickness
- 8. Many worries, often seems worried
- 13. Often unhappy, down-hearted or tearful
- 16. Nervous or clingy in new situations, easily loses confidence



- 24. Many fears, easily scared

## 2. Conduct Problems Scale:

- 5. Often has temper tantrums or hot tempers
- 7. Generally obedient, usually does what adults request (reverse scored)
- 12. Often fights with other children or bullies them
- 18. Often lies or cheats
- 22. Steals from home, school or elsewhere

## 3. Hyperactivity Scale:

- 2. Restless, overactive, cannot stay still for long
- 10. Constantly fidgeting or squirming
- 15. Easily distracted, concentration wanders
- 21. Thinks things out before acting (reverse scored)
- 25. Sees tasks through to the end, good attention span (reverse scored)

## 4. Peer Problems Scale:

- 6. Rather solitary, tends to play alone
- 11. Has at least one good friend (reverse scored)
- 14. Generally liked by other children (reverse scored)
- 19. Picked on or bullied by other children
- 23. Gets on better with adults than with other children

## 5. Prosocial Scale:

- 1. Considerate of other people's feelings
- 4. Shares readily with other children (treats, toys, pencils etc.)
- 9. Helpful if someone is hurt, upset or feeling ill
- 17. Kind to younger children
- 20. Often volunteers to help others (parents, teachers, other children)

Note: Items marked as "reverse scored" are phrased positively, so their scores are reversed when calculating the total for that scale. In the chapter's main analysis, the four difficulty subscales are then reverse-coded once more so that higher values indicate more positive traits: Emotional Resilience, Good Conduct, Focused Behaviour, and Positive Peer Relationships.

## 6.7 Translog elasticity-of-substitution formula and interpretation

This appendix documents the local Hicks elasticity-of-substitution expression for the two-input translog specification used in Section 4 and explains why that object is not used as a headline result in the main text. Here the Hicks elasticity is a local summary of how easily one input can be traded off against the other while holding output constant. Its computation is a standard diagnostic for translog specifications, but in this dataset the resulting quantity is substantially more fragile than the first-order translog quantities used in the main text. That fragility changes how the elasticity should be interpreted, but it does not change the chapter’s defended findings, which rest on the linear interaction results and the more stable first-order translog quantities.

For the translog specification

$$\ln Y = \alpha_0 + \alpha_1 \ln C + \alpha_2 \ln N + \frac{1}{2}\gamma_1(\ln C)^2 + \frac{1}{2}\gamma_2(\ln N)^2 + \gamma_{12} \ln C \ln N, \quad (3)$$

The output elasticities at a given evaluation point are

$$OE_C = \alpha_1 + \gamma_1 \ln C + \gamma_{12} \ln N, \quad OE_N = \alpha_2 + \gamma_2 \ln N + \gamma_{12} \ln C. \quad (4)$$

Using these local elasticities, the two-input Hicks elasticity of substitution is

$$\sigma = \frac{OE_C + OE_N}{(OE_C + OE_N) + 2\gamma_{12} - \gamma_2 \left( \frac{OE_C}{OE_N} \right) - \gamma_1 \left( \frac{OE_N}{OE_C} \right)}, \quad (5)$$

where  $\gamma_1$  and  $\gamma_2$  denote the own-curvature parameters on  $(\ln C)^2$  and  $(\ln N)^2$  in the underlying translog (sometimes written  $\gamma_{11}$  and  $\gamma_{22}$ ), and  $\gamma_{12}$  is the coefficient on the cross-term  $\ln C \cdot \ln N$ . The same  $\gamma_1$  and  $\gamma_2$  enter the output-elasticity expressions in equation (4) above.

Two points follow directly. First, this is a local object: it depends on the evaluation point through  $OE_C$  and  $OE_N$ , so it is not a single global parameter of the fitted surface. Second, the denominator can become small or unstable when one output elasticity is modest relative to the other or when the curvature terms interact unfavourably with those elasticities. In practice, that makes the local elasticity much more sensitive than the first-order quantities.

That sensitivity is exactly what appears in the data. Across the bootstrap exercises reported in Table 21, marginal products and output elasticities are locally regular in essentially every draw, while the local elasticity is often negative or extremely large in magnitude. I therefore treat the Hicks object as a fragile descriptive summary. In this setting, it does not operate as a stable structural parameter.

The substantive implication is narrow. This discussion strengthens the case for not

relying on a translog elasticity-of-substitution estimate as the headline result of the chapter. It does not overturn the main empirical pattern: the linear interaction terms remain predominantly negative, the SDQ-based interaction is the most robust result, and the translog remains useful mainly as a local curvature check. In this chapter, it does not support a strong claim about how easily cognitive and noncognitive inputs can be traded off.

## 6.8 Selection and attrition diagnostics

Table 16 compares baseline characteristics of retained versus lost observations at two selection margins. The retained analytical sample is positively selected on prior achievement and socioeconomic status, but effect sizes are moderate (Cohen’s  $d$  is about 0.35 or smaller throughout). Table 17 reports inverse-probability-weighted OLS interaction coefficients: the primary SDQ Focused Behaviour  $\times$  Cognition interaction is robust to selection reweighting, with coefficient changes of roughly 3–11%.

## 6.9 Additional robustness checks

This appendix collects supplementary robustness tables referenced in the main text, organised by the concern each addresses. Table 18 tests the linear interaction result under outcome scaling alternatives (z-scored vs. log). Table 21 reports a translog fragility diagnostic: under the Hicks elasticity-of-substitution formula reported in Appendix 6.7, the first-order translog objects (marginal products and output elasticities) remain regular, but the local ES object is unstable across a substantial share of bootstrap draws. The table is retained as a quantitative companion to the fragility discussion in the main text and Appendix 6.7; it is not a “well-behaved production function” diagnostic. Tables 22–23 gauge how much selection on unobservables would be needed to eliminate the key linear interaction effects. Tables 24 and 25 simulate school-level clustering to test whether significance survives without access to school identifiers.

The more conservative  $R^2_{\max}$  in Table 23 also overturns one TIPI cell relative to the headline Oster table. The Maths Openness  $\times$  Cog interaction is marked “Yes” in Table 22 ( $\delta^* = 2.89$ , identified set  $[0.0055, 0.0084]$ , both positive) but loses its robustness designation under the more conservative bound ( $\delta^* = 0.96 < 1$ , identified set  $[-0.0004, 0.0084]$ , including zero). The reversal is consistent with the broader pattern that TIPI-based interactions are more sensitive to the assumed  $R^2$  ceiling than the SDQ-based interactions, and reinforces the chapter’s reading of the TIPI cells as fragile relative to the SDQ Focused Behaviour result.

Table 16: Baseline balance: retained vs. lost observations at two selection margins. Panel A compares Wave 1 characteristics of the full GUI cohort ( $N = 8,568$ ) by retention in the analytical sample ( $N = 5,631$ ). Panel B compares the analytical sample by retention in the restricted (complete-controls) sample ( $N \approx 3,786$ ). Cohen’s  $d$  is the standardised mean difference (positive = retained group higher).

| Variable                                                                                  | Retained |         | Lost  |         | $d$    | $p$   |
|-------------------------------------------------------------------------------------------|----------|---------|-------|---------|--------|-------|
|                                                                                           | Mean     | SD      | Mean  | SD      |        |       |
| <i>Panel A: Full cohort <math>\rightarrow</math> analytical sample</i>                    |          |         |       |         |        |       |
| Reading (% correct, W1)                                                                   | 73.10    | (19.86) | 65.88 | (21.79) | +0.351 | 0.000 |
| Maths (% correct, W1)                                                                     | 58.82    | (20.44) | 51.76 | (21.28) | +0.341 | 0.000 |
| SDQ Emotional symptoms (W1)                                                               | 1.96     | (1.94)  | 2.11  | (2.03)  | -0.072 | 0.002 |
| SDQ Conduct problems (W1)                                                                 | 1.18     | (1.38)  | 1.39  | (1.51)  | -0.144 | 0.000 |
| SDQ Hyperactivity (W1)                                                                    | 2.87     | (2.37)  | 3.21  | (2.51)  | -0.143 | 0.000 |
| SDQ Peer problems (W1)                                                                    | 1.09     | (1.41)  | 1.24  | (1.47)  | -0.101 | 0.000 |
| SDQ Total difficulties (W1)                                                               | 7.11     | (4.86)  | 7.94  | (5.32)  | -0.167 | 0.000 |
| Income quintile (equiv., W1)                                                              | 3.43     | (1.34)  | 3.11  | (1.43)  | +0.230 | 0.000 |
| <i>Panel B: Analytical <math>\rightarrow</math> restricted (complete controls) sample</i> |          |         |       |         |        |       |
| Reading (% correct, W1)                                                                   | 74.46    | (19.23) | 70.30 | (20.84) | +0.210 | 0.000 |
| Maths (% correct, W1)                                                                     | 60.17    | (20.08) | 56.04 | (20.88) | +0.203 | 0.000 |
| SDQ Emotional res. (W1)                                                                   | 8.12     | (1.89)  | 7.86  | (2.04)  | +0.136 | 0.000 |
| SDQ Good conduct (W1)                                                                     | 8.86     | (1.35)  | 8.73  | (1.45)  | +0.098 | 0.001 |
| SDQ Focused behav. (W1)                                                                   | 7.19     | (2.35)  | 7.02  | (2.41)  | +0.070 | 0.014 |
| SDQ Pos. peers (W1)                                                                       | 8.98     | (1.36)  | 8.75  | (1.50)  | +0.159 | 0.000 |
| Maths points (JC outcome)                                                                 | 9.75     | (1.68)  | 9.30  | (1.81)  | +0.259 | 0.000 |
| English points (JC outcome)                                                               | 10.25    | (1.26)  | 9.94  | (1.45)  | +0.232 | 0.000 |
| SDQ Focused behav. (W2)                                                                   | 7.66     | (2.21)  | 7.38  | (2.36)  | +0.124 | 0.000 |
| TIPi Conscient. (W2)                                                                      | 4.36     | (2.06)  | 4.26  | (2.10)  | +0.047 | 0.097 |
| PCG Education (W2)                                                                        | 4.02     | (1.23)  | 3.84  | (1.26)  | +0.144 | 0.000 |

*Note:*  $d$  is Cohen’s standardised mean difference (positive = retained group higher).  $p$ -values from Welch’s two-sample  $t$ -test. Panel A uses the raw Wave 1 SDQ difficulty-subscale names from the GUI documentation; Panel B uses the reverse-coded positive-trait names used in the chapter’s main analysis. The sign switch across panels therefore reflects a naming convention, not a reversal in the selection pattern.

Table 17: IPW sensitivity: OLS interaction coefficients with and without inverse-probability weighting for selection into the restricted (complete-controls) sample. Weights are estimated from a logit of sample retention on Wave 1 baseline characteristics, trimmed at the 1st/99th percentiles, and applied to the controls-only linear specification so that the selection-correction channel can be read separately from the later Wave 1-baseline adjustment.

| Outcome | Interaction                 | Unweighted |          | IPW-weighted |          | Change (%) |
|---------|-----------------------------|------------|----------|--------------|----------|------------|
|         |                             | Coef       | SE       | Coef         | SE       |            |
| Maths   | Emot. Res. $\times$ Cog     | -0.0430*   | (0.0216) | -0.0399+     | (0.0224) | +7.2%      |
| Maths   | Good Conduct $\times$ Cog   | 0.0172     | (0.0240) | 0.0186       | (0.0249) | +8.2%      |
| Maths   | Focused Behav. $\times$ Cog | -0.0828*** | (0.0231) | -0.0739**    | (0.0246) | +10.8%     |
| Maths   | Pos. Peer Rel. $\times$ Cog | -0.0199    | (0.0192) | -0.0234      | (0.0202) | -17.5%     |
| English | Emot. Res. $\times$ Cog     | -0.0263    | (0.0208) | -0.0240      | (0.0218) | +8.9%      |
| English | Good Conduct $\times$ Cog   | 0.0324     | (0.0240) | 0.0322       | (0.0252) | -0.5%      |
| English | Focused Behav. $\times$ Cog | -0.0979*** | (0.0232) | -0.1004***   | (0.0250) | -2.6%      |
| English | Pos. Peer Rel. $\times$ Cog | -0.0379+   | (0.0198) | -0.0389+     | (0.0207) | -2.8%      |

*Note:* IPW weights are the inverse of  $\Pr(\text{in restricted sample} \mid \text{Wave 1 baselines})$ , estimated by logit on the full analytical sample ( $N = 5,631$ ) and applied to the restricted sample ( $N \approx 3,786$ ). Weights trimmed at 1st/99th percentiles and normalised to sum to  $N$ . The SDQ interaction labels use the reverse-coded positive-trait names from the main text, so “Focused Behaviour  $\times$  Cog” is the focal interaction. HC1 robust SEs. \*\*\* $p < 0.001$ , \*\* $p < 0.01$ , \* $p < 0.05$ , + $p < 0.1$ .

Table 18: Robustness: second-order polynomial with z-scored outcomes vs. log outcomes. The cross-term coefficient ( $\hat{\beta}_{C \times N}$ ) captures the interaction between cognitive and noncognitive inputs. A negative cross-term indicates a negative cross-effect in the levels-based specification, and the table checks whether that sign pattern is preserved when the outcome is standardised. This compares the result with the logged-outcome specification.

| Outcome | Noncog     | Subgroup    | $\hat{\beta}_{C \times N}$ (z-score) | 95% CI           | $\hat{\beta}_{C \times N}$ (log) | Sign match |
|---------|------------|-------------|--------------------------------------|------------------|----------------------------------|------------|
| Maths   | SDQ (F.B.) | Full sample | -0.0362***                           | [-0.058, -0.015] | -0.0105***                       | ✓          |
| Maths   | SDQ (F.B.) | Boys        | -0.0317*                             | [-0.062, -0.001] | -0.0096**                        | ✓          |
| Maths   | SDQ (F.B.) | Girls       | -0.0377*                             | [-0.069, -0.006] | -0.0110**                        | ✓          |
| Maths   | TIPI (C.)  | Full sample | -0.0219*                             | [-0.041, -0.004] | -0.0054*                         | ✓          |
| Maths   | TIPI (C.)  | Boys        | -0.0389**                            | [-0.066, -0.009] | -0.0097**                        | ✓          |
| Maths   | TIPI (C.)  | Girls       | -0.0056                              | [-0.034, 0.023]  | -0.0017                          | ✓          |
| English | SDQ (F.B.) | Full sample | -0.0559***                           | [-0.082, -0.030] | -0.0096***                       | ✓          |
| English | SDQ (F.B.) | Boys        | -0.0477*                             | [-0.089, -0.008] | -0.0087**                        | ✓          |
| English | SDQ (F.B.) | Girls       | -0.0664***                           | [-0.106, -0.027] | -0.0105***                       | ✓          |
| English | TIPI (C.)  | Full sample | -0.0177                              | [-0.042, 0.007]  | -0.0031                          | ✓          |
| English | TIPI (C.)  | Boys        | -0.0267                              | [-0.065, 0.008]  | -0.0046                          | ✓          |
| English | TIPI (C.)  | Girls       | -0.0035                              | [-0.037, 0.032]  | -0.0008                          | ✓          |

*Note:* F.B. = Focused Behaviour; C. = Conscientiousness. Second-order polynomial:  $z(Y) = \alpha + \beta_1 C + \beta_2 N + \beta_3 C^2 + \beta_4 N^2 + \beta_5 C \times N$ , where  $C$  = cognition (PC, standardised) and  $N$  = noncognitive measure (standardised). Z-score specification uses the standardised outcome (mean 0, SD 1); log specification uses  $\ln(Y)$ . Absolute z-score coefficients are larger because  $z(Y)$  and  $\ln(Y)$  are on different outcome scales. For this robustness check, the relevant comparison is sign preservation, not numerical equality. 95% CIs are percentile bootstrap (1,000 replications). HC1 robust SEs for point estimates. “Sign match” indicates whether the cross-term has the same sign in both specifications. \*\*\* $p < 0.001$ , \*\* $p < 0.01$ , \* $p < 0.05$ , + $p < 0.1$ .

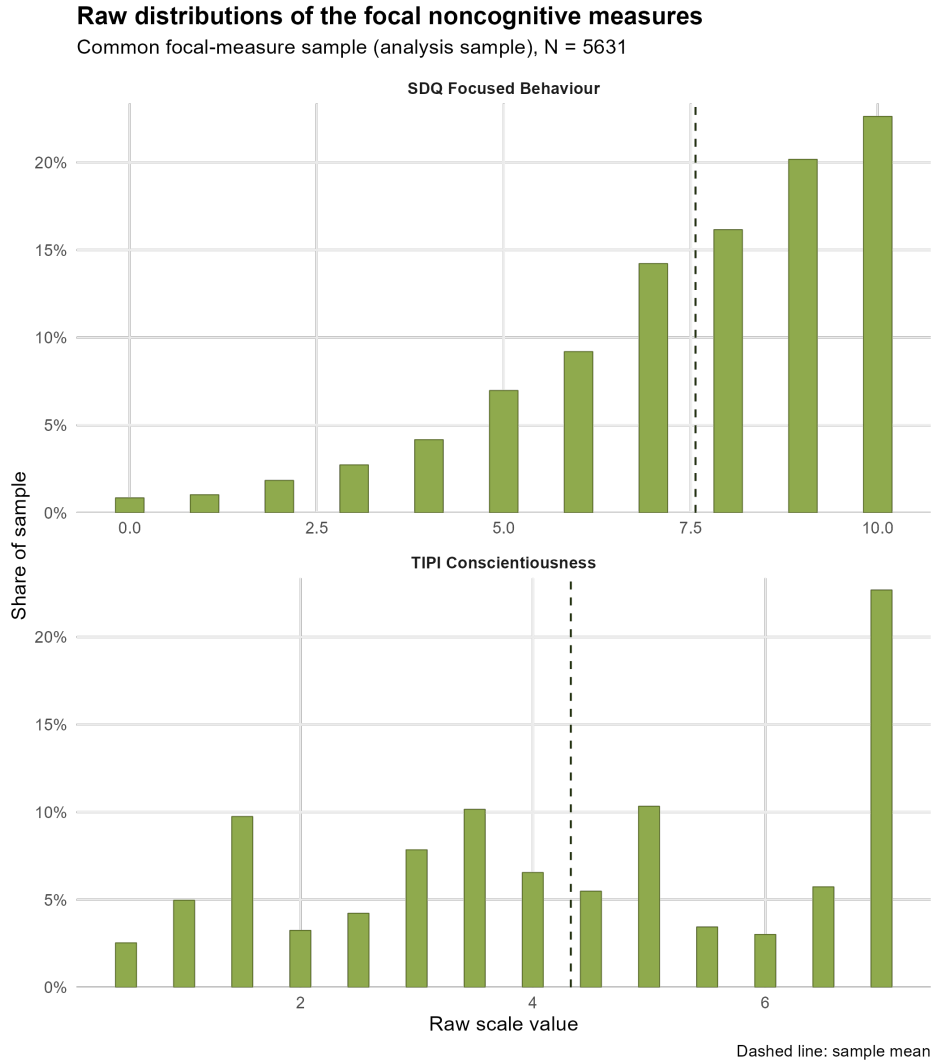


Figure 3: Raw distributions of the focal noncognitive measures on the common focal-measure sample ( $N = 5,631$ ). Bars report the share of the sample at each observed scale value, and the dashed vertical line marks the sample mean within each panel. The SDQ Focused Behaviour measure is more concentrated near the upper end of its 0–10 scale, while TIPI Conscientiousness is more dispersed across its 0.5–7 range with a visible upper-bound pile-up. The figure is reported as a descriptive diagnostic of scale use and distributional shape. It is not intended as a formal measurement model.

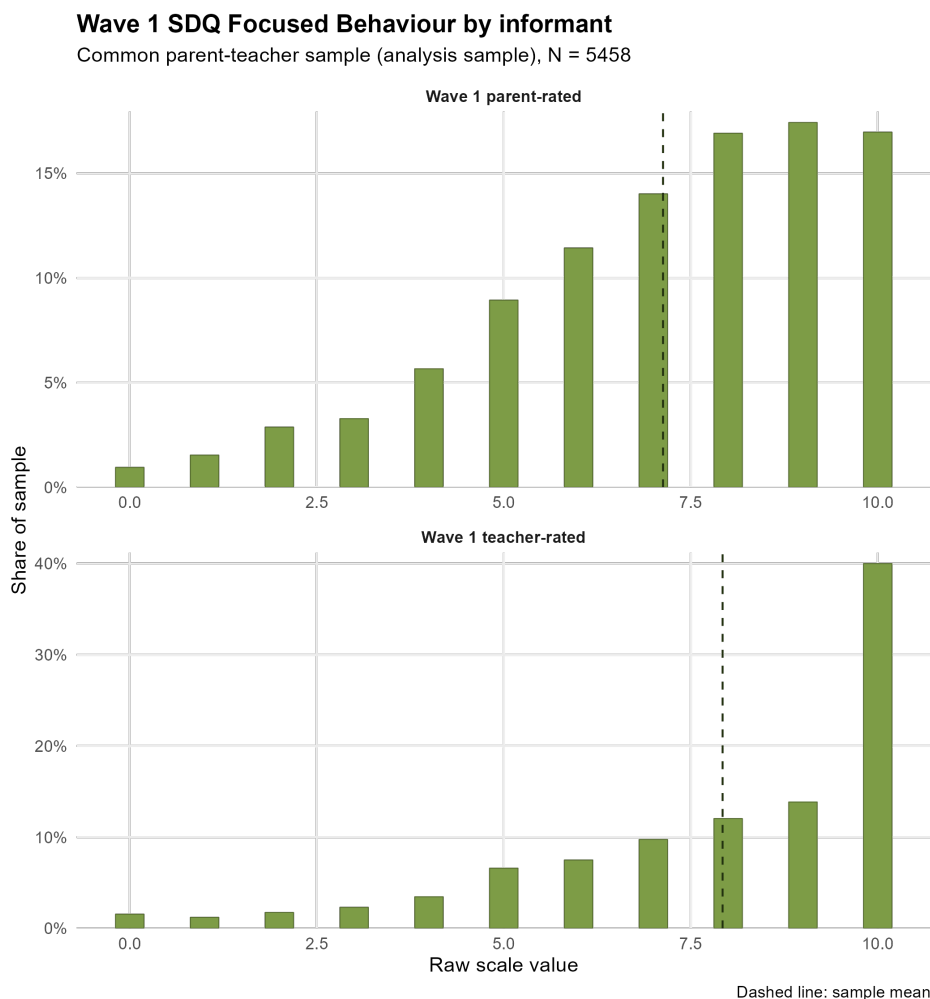


Figure 4: Wave 1 cross-informant raw distributions for SDQ Focused Behaviour on the common parent-teacher sample ( $N = 5,458$ ). Bars report the share of the sample at each observed scale value, and the dashed vertical line marks the sample mean within each panel. The teacher-rated Wave 1 measure shows a much stronger ceiling pile-up at the upper bound than the parent-rated Wave 1 measure, which helps clarify how scale use differs across informants even before the interaction models are estimated.

Table 19: Direct same-sample test of whether the focal SDQ and TIPI interaction coefficients differ. Each row estimates a pooled model including both SDQ Focused Behaviour and TIPI Conscientiousness, together with their interactions with cognition, and tests the null that the two interaction coefficients are equal.

| Outcome | Specification      | $N$  | $\hat{\beta}_{SDQ \times Cog}$ | $\hat{\beta}_{TIPI \times Cog}$ | Difference | SE(diff) | p-value |
|---------|--------------------|------|--------------------------------|---------------------------------|------------|----------|---------|
| Maths   | Controls only      | 3786 | -0.0681                        | -0.0073                         | -0.0609    | 0.0275   | 0.0269  |
| English | Controls only      | 3786 | -0.0758                        | -0.0000                         | -0.0757    | 0.0261   | 0.0037  |
| Maths   | + Wave 1 baselines | 3711 | -0.0787                        | -0.0068                         | -0.0719    | 0.0278   | 0.0097  |
| English | + Wave 1 baselines | 3711 | -0.0838                        | 0.0020                          | -0.0858    | 0.0260   | 0.0010  |

*Note:* HC1-robust inference. “Controls only” uses the common restricted sample with cognition, the two focal noncognitive inputs, SES controls, and school controls. “+ Wave 1 baselines” adds age-9 reading and maths scores and the four parent-reported Wave 1 SDQ subscales. Because this pooled model includes both focal noncognitive inputs and both interactions simultaneously, the SDQ coefficient reported here need not equal the single-instrument SDQ coefficient reported in the main-text tables. The reported p-value is for the null  $H_0 : \beta_{SDQ \times Cog} = \beta_{TIPI \times Cog}$ , computed from the robust covariance matrix of the pooled model.

Table 20: TIPI Conscientiousness W2 contamination sensitivity. Focal Conscientiousness  $\times$  Cognition interaction in the Wave 1-baseline-controlled Model 4 specification, under three trim variants on the GUI-supplied W2 PCG TIPI Conscientiousness input.

| Trim                       | Outcome | $N$   | Coef    | SE     | $p$   |
|----------------------------|---------|-------|---------|--------|-------|
| Baseline                   | Maths   | 3,708 | -0.0144 | 0.0154 | 0.349 |
| Baseline                   | English | 3,708 | -0.0149 | 0.0140 | 0.286 |
| Visible trim ( $< 1.0$ )   | Maths   | 3,626 | -0.0163 | 0.0152 | 0.284 |
| Visible trim ( $< 1.0$ )   | English | 3,626 | -0.0176 | 0.0143 | 0.216 |
| Ambig. trim ( $\leq 3.5$ ) | Maths   | 2,149 | -0.0133 | 0.0194 | 0.493 |
| Ambig. trim ( $\leq 3.5$ ) | English | 2,149 | +0.0108 | 0.0173 | 0.533 |

*Notes:* Each row reports the Conscientiousness W2 PCG  $\times$  Cognition interaction coefficient from the preferred Model 4 specification (Wave 2 cognition PC, all five W2 TIPI traits, all five cognition  $\times$  TIPI interactions, SES and school controls, and six Wave 1 baselines), mirroring the Maths\_TIPI / English\_TIPI rows of Table 11; only the trim on the focal Conscientiousness input changes across rows. “Visible trim” drops the 82 cases below the documented 1.0 Likert minimum. “Ambiguous trim” drops the 1,559 cases at or below 3.5, the conservative upper bound on the range that could result from a  $(0 + x)/2$  missing-item mixture given items in  $[1, 7]$ ; it is reported for completeness but is too aggressive to be informative because it removes the entire low end of the distribution. HC1 robust standard errors.



Table 21: Translog fragility diagnostic. Bootstrap distribution of regularity outcomes at the sample-mean evaluation point under the Hicks elasticity-of-substitution formula. The first-order objects (marginal products of cognitive and noncognitive inputs) are regular in essentially every draw, while the second-order local ES object is often negative or extreme in magnitude across every specification. The table is reported as a quantitative companion to the fragility discussion in the main text; it is *not* a “well-behaved production function” diagnostic. 1,000 bootstrap replications.

| Sample | Outcome | Noncog | Shift | % $MP_C < 0$ | % $MP_N < 0$ | % $ES < 0$ | % $ ES  > 10$ |
|--------|---------|--------|-------|--------------|--------------|------------|---------------|
| Full   | Maths   | SDQ    | 0.5   | 0.0          | 0.0          | 7.2        | 24.0          |
| Full   | Maths   | SDQ    | 1.0   | 0.0          | 0.0          | 54.7       | 49.4          |
| Full   | Maths   | TIPI   | 0.5   | 0.0          | 0.0          | 65.1       | 33.1          |
| Full   | Maths   | TIPI   | 1.0   | 0.0          | 0.0          | 90.5       | 9.7           |
| Full   | English | SDQ    | 0.5   | 0.0          | 0.0          | 41.2       | 38.4          |
| Full   | English | SDQ    | 1.0   | 0.0          | 0.0          | 81.0       | 27.4          |
| Full   | English | TIPI   | 0.5   | 0.0          | 0.0          | 61.1       | 26.6          |
| Full   | English | TIPI   | 1.0   | 0.0          | 0.0          | 83.6       | 11.6          |
| Boys   | Maths   | SDQ    | 0.5   | 0.0          | 0.0          | 7.3        | 18.9          |
| Boys   | Maths   | TIPI   | 0.5   | 0.0          | 0.0          | 50.4       | 20.5          |
| Boys   | English | SDQ    | 0.5   | 0.0          | 0.0          | 44.4       | 38.6          |
| Boys   | English | TIPI   | 0.5   | 0.0          | 0.0          | 66.2       | 11.2          |
| Girls  | Maths   | SDQ    | 0.5   | 0.0          | 0.0          | 31.1       | 32.4          |
| Girls  | Maths   | TIPI   | 0.5   | 0.0          | 0.0          | 69.3       | 27.1          |
| Girls  | English | SDQ    | 0.5   | 0.0          | 0.0          | 74.5       | 17.9          |
| Girls  | English | TIPI   | 0.5   | 0.0          | 0.1          | 37.5       | 15.3          |

*Notes:* Each row reports the share of 1,000 bootstrap replications, evaluated at the sample-mean reference point after centering the log inputs, in which the indicated quantity violates a regularity condition.  $MP_C$  and  $MP_N$  are the marginal products of cognitive and noncognitive inputs from the fitted translog. ES is the two-input Hicks elasticity of substitution,  $\sigma = (OE_C + OE_N)/[(OE_C + OE_N) + 2\gamma_{12} - \gamma_2(OE_C/OE_N) - \gamma_1(OE_N/OE_C)]$ . The marginal products are positive in essentially every bootstrap draw across every specification, so the fitted production surface itself is locally regular. The local ES object, by contrast, is negative in 7–91% of draws and exceeds  $|10|$  in another 10–49% of draws, with the SDQ-Maths shift 0.5 cell the least unstable and the TIPI specifications the most. This is the empirical counterpart to the argument in the main text: the local Hicks elasticity combines small denominators with curvature ratios that are very sensitive to the magnitude of the noncognitive output elasticity, and is not a structural parameter that can be precisely estimated in this setting. My defended findings rest on the linear interaction patterns reported in Section 5 and the cross-informant robustness in Section 5, not on the local ES object summarised here.

Table 22: Oster (2019) bounds for omitted variable bias. For each cognitive–noncognitive interaction,  $\delta^*$  reports the proportional degree of selection on unobservables (relative to observables) required to drive the coefficient to zero.  $\delta^* > 1$  indicates that unobservables would need to be more correlated with the interaction than the full set of included controls.  $\beta^*(\delta = 1)$  is the bias-adjusted coefficient under equal selection.  $R_{\max}^2 = 1.3 \times R_{\text{full}}^2$  (Oster’s recommended heuristic). Full model includes W2 cognitive (PC), noncognitive subscales, SES/school controls, Wave 1 baselines, and all cognition  $\times$  noncognition interactions (HC1 SEs).

| Outcome | Interaction                              | $\hat{\beta}$        | SE       | $\delta^*$ | $\beta^*(\delta = 1)$ | $[\beta, \beta^*]$ | Robust           |
|---------|------------------------------------------|----------------------|----------|------------|-----------------------|--------------------|------------------|
| Maths   | Emotional Resilience $\times$ Cog        | -0.0172              | (0.0162) | 0.58       | 0.0127                | [-0.0172, 0.0127]  |                  |
| Maths   | Good Conduct $\times$ Cog                | 0.0088               | (0.0176) | -0.12      | 0.0835                | [0.0088, 0.0835]   | Yes <sup>†</sup> |
| Maths   | Focused Behaviour $\times$ Cog           | -0.0517**            | (0.0168) | -6.56      | -0.0596               | [-0.0596, -0.0517] | Yes <sup>†</sup> |
| Maths   | Positive Peer Relationships $\times$ Cog | -0.0097              | (0.0145) | 0.22       | 0.0340                | [-0.0097, 0.0340]  |                  |
| English | Emotional Resilience $\times$ Cog        | -0.0072              | (0.0147) | 0.37       | 0.0125                | [-0.0072, 0.0125]  |                  |
| English | Good Conduct $\times$ Cog                | 0.0196               | (0.0171) | -0.37      | 0.0731                | [0.0196, 0.0731]   | Yes <sup>†</sup> |
| English | Focused Behaviour $\times$ Cog           | -0.0620***           | (0.0161) | 4.84       | -0.0492               | [-0.0620, -0.0492] | Yes              |
| English | Positive Peer Relationships $\times$ Cog | -0.0198              | (0.0141) | 0.72       | 0.0075                | [-0.0198, 0.0075]  |                  |
| Maths   | Agreeableness $\times$ Cog               | -0.0119              | (0.0162) | 0.31       | 0.0260                | [-0.0119, 0.0260]  |                  |
| Maths   | Conscientiousness $\times$ Cog           | -0.0144              | (0.0154) | 0.45       | 0.0180                | [-0.0144, 0.0180]  |                  |
| Maths   | Emotional Stability $\times$ Cog         | -0.0614***           | (0.0154) | -1.74      | -0.0967               | [-0.0967, -0.0614] | Yes <sup>†</sup> |
| Maths   | Extraversion $\times$ Cog                | 0.0036               | (0.0150) | -0.28      | 0.0167                | [0.0036, 0.0167]   | Yes <sup>†</sup> |
| Maths   | Openness $\times$ Cog                    | 0.0084               | (0.0155) | 2.89       | 0.0055                | [0.0055, 0.0084]   | Yes              |
| English | Agreeableness $\times$ Cog               | -0.0094              | (0.0145) | 0.84       | 0.0018                | [-0.0094, 0.0018]  |                  |
| English | Conscientiousness $\times$ Cog           | -0.0149              | (0.0140) | 1.02       | -0.0002               | [-0.0149, -0.0002] | Yes              |
| English | Emotional Stability $\times$ Cog         | -0.0201              | (0.0142) | 1.81       | -0.0090               | [-0.0201, -0.0090] | Yes              |
| English | Extraversion $\times$ Cog                | -0.0013              | (0.0131) | 0.08       | 0.0152                | [-0.0013, 0.0152]  |                  |
| English | Openness $\times$ Cog                    | -0.0230 <sup>+</sup> | (0.0135) | 4.57       | -0.0180               | [-0.0230, -0.0180] | Yes              |

Note:  $\delta^*$  is the proportional selection on unobservables (relative to observables) needed to drive  $\hat{\beta} \rightarrow 0$  (Oster, 2019). Positive  $\delta^* > 1$ : unobservables would need to exceed observables in explanatory power. Negative  $\delta^*$ : adding controls moves the coefficient *away* from zero (bias-amplifying), marked <sup>†</sup>. In these cases the Oster framework cannot bound the coefficient toward zero through the observable-to-unobservable proportionality assumption, because the implied extrapolation goes the wrong way. The “Yes<sup>†</sup>” designation in the Robust column therefore reflects the direction of selection (i.e. the identified set lies entirely on one side of zero), not a passed threshold on the size of unobservable selection that would be required to eliminate the effect through a different selection mechanism.  $\beta^*(\delta = 1)$  is the identified-set bound under equal selection.  $R_{\max}^2 = 1.3 \times R_{\text{full}}^2$ . \*\*\* $p < 0.001$ , \*\* $p < 0.01$ , \* $p < 0.05$ , <sup>+</sup> $p < 0.1$ .

Table 23: Oster (2019) bounds: sensitivity to  $R_{\max}^2$ . Same specifications as Table 22 but with  $R_{\max}^2 = \min(2.2 \times R_{\text{full}}^2, 1)$ , a more conservative upper bound on attainable  $R^2$ .

| Outcome | Interaction                              | $\hat{\beta}$ | $R_{\max}^2$ | $\delta^*$ | $\beta^*(\delta=1)$ | Robust           |
|---------|------------------------------------------|---------------|--------------|------------|---------------------|------------------|
| Maths   | Emotional Resilience $\times$ Cog        | -0.0172       | 1.000        | 0.19       | 0.0713              |                  |
| Maths   | Good Conduct $\times$ Cog                | 0.0088        | 1.000        | -0.04      | 0.2298              | Yes <sup>†</sup> |
| Maths   | Focused Behaviour $\times$ Cog           | -0.0517       | 1.000        | -2.22      | -0.0750             | Yes <sup>†</sup> |
| Maths   | Positive Peer Relationships $\times$ Cog | -0.0097       | 1.000        | 0.07       | 0.1198              |                  |
| English | Emotional Resilience $\times$ Cog        | -0.0072       | 0.873        | 0.09       | 0.0713              |                  |
| English | Good Conduct $\times$ Cog                | 0.0196        | 0.873        | -0.09      | 0.2336              | Yes <sup>†</sup> |
| English | Focused Behaviour $\times$ Cog           | -0.0620       | 0.873        | 1.21       | -0.0107             | Yes              |
| English | Positive Peer Relationships $\times$ Cog | -0.0198       | 0.873        | 0.18       | 0.0896              |                  |
| Maths   | Agreeableness $\times$ Cog               | -0.0119       | 1.000        | 0.10       | 0.1026              |                  |
| Maths   | Conscientiousness $\times$ Cog           | -0.0144       | 1.000        | 0.15       | 0.0834              |                  |
| Maths   | Emotional Stability $\times$ Cog         | -0.0614       | 1.000        | -0.58      | -0.1680             | Yes <sup>†</sup> |
| Maths   | Extraversion $\times$ Cog                | 0.0036        | 1.000        | -0.09      | 0.0432              | Yes <sup>†</sup> |
| Maths   | Openness $\times$ Cog                    | 0.0084        | 1.000        | 0.96       | -0.0004             |                  |
| English | Agreeableness $\times$ Cog               | -0.0094       | 0.860        | 0.21       | 0.0354              |                  |
| English | Conscientiousness $\times$ Cog           | -0.0149       | 0.860        | 0.25       | 0.0438              |                  |
| English | Emotional Stability $\times$ Cog         | -0.0201       | 0.860        | 0.45       | 0.0242              |                  |
| English | Extraversion $\times$ Cog                | -0.0013       | 0.860        | 0.02       | 0.0644              |                  |
| English | Openness $\times$ Cog                    | -0.0230       | 0.860        | 1.14       | -0.0029             | Yes              |

Note: See Table 22 for definitions. <sup>†</sup>Bias-amplifying: controls move coefficient away from zero. The same direction-of-selection caveat applies as in Table 22: a “Yes<sup>†</sup>” designation reflects the sign of selection rather than a passed threshold.

## 6.10 Complementarity across developmental windows

This appendix documents the longer-horizon exploratory extensions mentioned in the main text. I considered two descriptive variants that replace Junior Certificate outcomes with Leaving Certificate points: first, the same Wave 2 cognitive and noncognitive inputs predicting Leaving Certificate outcomes at Wave 4; second, a near-contemporaneous Wave 3-to-Leaving-Certificate design that also changes the cognitive test battery.

I re-estimated those longer-horizon translog extensions under the same Hicks formula used elsewhere in the chapter. The first-order objects remained broadly sensible, but the local elasticity summaries continued to be too fragile and evaluation-point-sensitive to support a stable longer-horizon nonlinear claim. The defended result of the chapter therefore remains the Junior-Certificate-based linear interaction pattern, with the translog retained only as a local curvature check in the main specification.

If the Leaving Certificate question is pursued further, it would be better handled as a dedicated longer-horizon exercise in which the outcome horizon varies without simultaneously changing the nonlinear summary object on which the interpretation depends.

Table 24: Design-effect adjusted standard errors. Columns report the HC1 coefficient and standard error alongside the cluster-adjusted SE under the conservative scenario ( $\bar{m} = 8$ ,  $\rho = 0.15$ ,  $\text{DEFF} = 2.05$ ). This inflates SEs by a factor of  $\sqrt{2.05} \approx 1.43$ , simulating the effect of school-level clustering without requiring school identifiers. Full model includes W2 cognitive (PC), noncognitive subscales, SES/school controls, Wave 1 baselines, and all cognition $\times$ noncognition interactions.

| Outcome | Interaction                              | $\hat{\beta}$ | $\text{SE}_{\text{HC1}}$ | $\text{SE}_{\text{adj}}$ | $t_{\text{adj}}$ |
|---------|------------------------------------------|---------------|--------------------------|--------------------------|------------------|
| Maths   | Emotional Resilience $\times$ Cog        | -0.0172       | 0.0162                   | 0.0231                   | -0.74            |
| Maths   | Good Conduct $\times$ Cog                | 0.0088        | 0.0176                   | 0.0251                   | 0.35             |
| Maths   | Focused Behaviour $\times$ Cog           | -0.0517       | 0.0168                   | 0.0240                   | -2.15            |
| Maths   | Positive Peer Relationships $\times$ Cog | -0.0097       | 0.0145                   | 0.0208                   | -0.47            |
| English | Emotional Resilience $\times$ Cog        | -0.0072       | 0.0147                   | 0.0211                   | -0.34            |
| English | Good Conduct $\times$ Cog                | 0.0196        | 0.0171                   | 0.0245                   | 0.80             |
| English | Focused Behaviour $\times$ Cog           | -0.0620       | 0.0161                   | 0.0230                   | -2.69            |
| English | Positive Peer Relationships $\times$ Cog | -0.0198       | 0.0141                   | 0.0203                   | -0.98            |
| Maths   | Agreeableness $\times$ Cog               | -0.0119       | 0.0162                   | 0.0231                   | -0.52            |
| Maths   | Conscientiousness $\times$ Cog           | -0.0144       | 0.0154                   | 0.0221                   | -0.65            |
| Maths   | Emotional Stability $\times$ Cog         | -0.0614       | 0.0154                   | 0.0221                   | -2.78            |
| Maths   | Extraversion $\times$ Cog                | 0.0036        | 0.0150                   | 0.0214                   | 0.17             |
| Maths   | Openness $\times$ Cog                    | 0.0084        | 0.0155                   | 0.0222                   | 0.38             |
| English | Agreeableness $\times$ Cog               | -0.0094       | 0.0145                   | 0.0208                   | -0.45            |
| English | Conscientiousness $\times$ Cog           | -0.0149       | 0.0140                   | 0.0200                   | -0.74            |
| English | Emotional Stability $\times$ Cog         | -0.0201       | 0.0142                   | 0.0204                   | -0.98            |
| English | Extraversion $\times$ Cog                | -0.0013       | 0.0131                   | 0.0187                   | -0.07            |
| English | Openness $\times$ Cog                    | -0.0230       | 0.0135                   | 0.0193                   | -1.19            |

Note: Design effect  $\text{DEFF} = 1 + (\bar{m} - 1)\rho$  where  $\bar{m}$  is the average cluster size and  $\rho$  is the intra-class correlation.  $\text{SE}_{\text{adj}} = \text{SE}_{\text{HC1}} \times \sqrt{\text{DEFF}}$ . Parameters:  $\bar{m} = 8$  students per school (GUI sampling estimate),  $\rho = 0.15$  (OECD PISA 2015 between-school variance share for Ireland).

Table 25: Design-effect sensitivity: SDQ Focused Behaviour  $\times$  Cognition (primary noncognitive interaction) across intra-class correlation values.  $\bar{m} = 8$ . The interaction remains significant at the 5% level for both subjects across the ICC scenarios reported here ( $\rho \leq 0.20$ ). At  $\rho = 0.20$  the Maths  $t$ -statistic is  $-1.99$ , so significance for Maths would be lost at modestly higher intra-class correlations; English remains significant at the 5% level up to  $\rho \approx 0.40$ .

| Outcome | $\rho$ | DEFF | $\hat{\beta}$ | $\text{SE}_{\text{adj}}$ | $t_{\text{adj}}$ |
|---------|--------|------|---------------|--------------------------|------------------|
| Maths   | 0.05   | 1.35 | -0.0517       | 0.0195                   | -2.65            |
| Maths   | 0.10   | 1.70 | -0.0517       | 0.0219                   | -2.36            |
| Maths   | 0.15   | 2.05 | -0.0517       | 0.0240                   | -2.15            |
| Maths   | 0.20   | 2.40 | -0.0517       | 0.0260                   | -1.99            |
| English | 0.05   | 1.35 | -0.0620       | 0.0187                   | -3.32            |
| English | 0.10   | 1.70 | -0.0620       | 0.0210                   | -2.96            |
| English | 0.15   | 2.05 | -0.0620       | 0.0230                   | -2.69            |
| English | 0.20   | 2.40 | -0.0620       | 0.0249                   | -2.49            |

Note: Design effect  $\text{DEFF} = 1 + (\bar{m} - 1)\rho$ .  $\bar{m} = 8$ .  $\text{SE}_{\text{adj}} = \text{SE}_{\text{HC1}} \times \sqrt{\text{DEFF}}$ . Full model: Model 4 with Wave 1 baselines.